MEDICOOL 10.0 P6 R407C

10kW Gradient/Coldhead Compressor Water Chiller

User Manual

（New Edition, rev 12）

Revision History

Revision	Changing Description	Date
12	a, Re-arrange all chapters of old manual. b, Replaced 'purified' with 'de-ionized' in Table- 2 "Packing list of spare parts kit", item 9. c, Replaced 'purified' with 'de-ionized' in Table-4 "Specification", coolant specification.	Apr 22, 2009

Symbols Used on the MEDICOOL 10.0 P6 R407C

Chapters and sections in this instruction marked with the symbol particularly **Relevant to Safety**.
When it attached to the unit, this symbol draws attention to the relevant section of the instruction manual.

This symbol indicates **hazardous voltages** may be present.

A ***'NOTE'*** is used to call particular attention to a step of a procedure, which, if not strictly followed, could result in damage to, or destruction of equipment.

Content

Section 1 – General Information

1.1. Unpacking

The unit is packed and transported in a special carton. The package dimension of the unit is 1620x870x1500 mm^3, gross weight is around 340kg (750pound). The package dimension of spare parts kit is 800x700x870 mm^3, gross weight is around 178.5kg (394pound).

1. Use a pry bar to open top cover of the crate, remove it and place in an area clear of the uncrating area. (Fig.- 1).

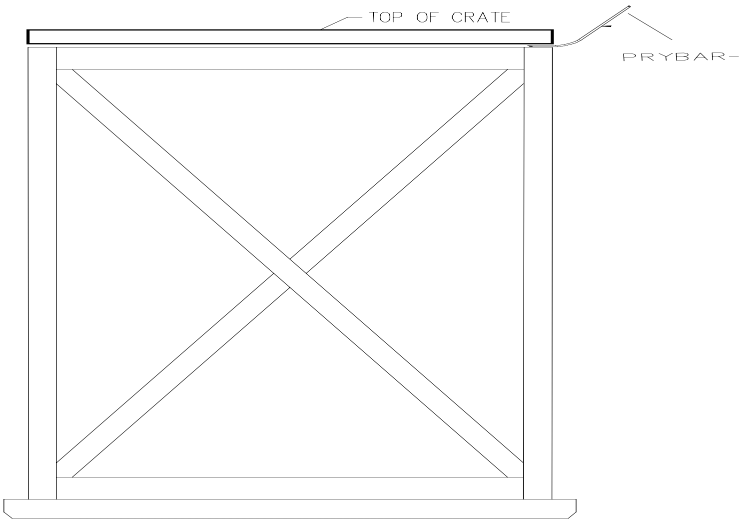


Fig.- 1 Remove top of the crate

2. Use a pry bar carefully remove the plate of crate around the unit, and place in an area clear of the uncrating area. (Fig.- 2)

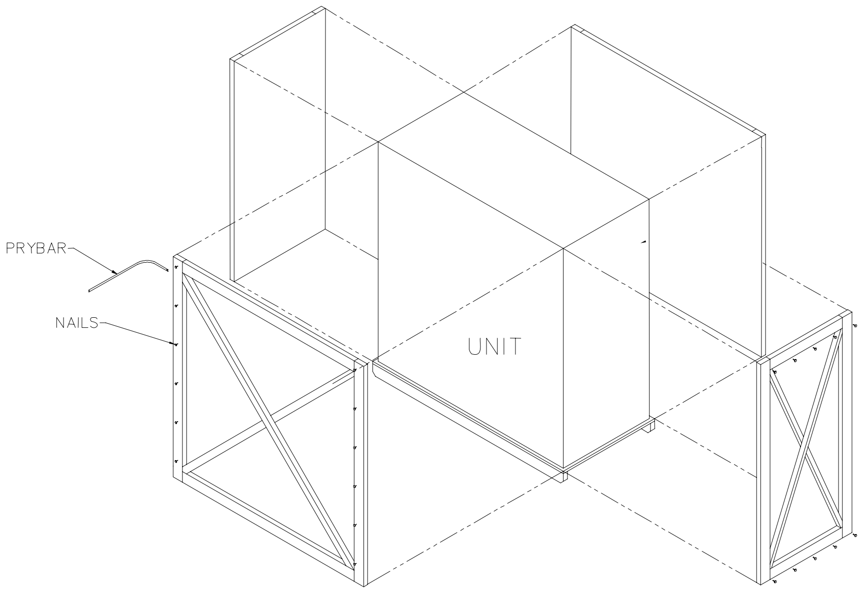


Fig.- 2 Remove other parts of the crate

3. CAREFULLY remove plastic protector.

4. Visual check the equipment for any damage during transportation. Contact AIRSYS and/or the forwarder as soon as possible if any damage is found. Air-sys can be reached by the information at the bottom of each page on the manual.

5. Prepare two lifting straps with hoisting eyes at both ends, approximately 5.08 cm (2 inches) wide and 3.05 m (10 feet) long. Place the two straps as Fig.-3, create a basket lift by pushing one end under unit and up opposite side until eyes are even (Fig.- 3), place straps 10.5cm(4.13 inches) from side.

6. Let the fork of the forklift through the eyes of the straps, like Fig.- 3. Attach a C-clamp, or similar clamping device, for each fork, as Fig.- 4, to prevent straps from sliding off.

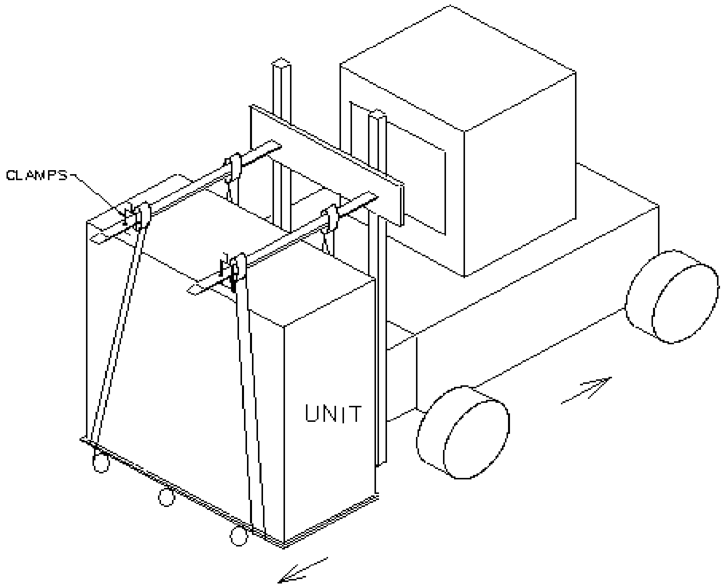


Fig.- 3 Unit lifting

7. Before move the unit to working place, try to lift the unit a little to see whether the two straps need to be adjusted to balance the unit from toppled over.

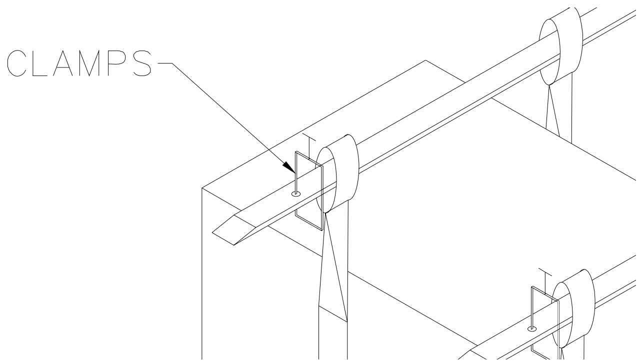


Fig.- 4 C-clamp placement

Keep the carton and all packing materials until the unit is completely assembled and working properly. Set up and run the unit immediately to confirm proper operation. If the unit is damaged or does not operate properly, contact the company who deliver the unit to your place, and file a damage claim. Inform GEHC or to whom your unit was purchased.

1.2. Service and Technical Support

Customer satisfaction is our highest priority. Please contact us immediately for technical assistance whenever you have questions or concerns. We can be reached by information listed in Table-1 or contact information at the bottom of each page.

Please address information listed below before you call service or technical support,

– Model and Serial Number
– Voltage (from back panel label)
– Date of purchase and your purchase order number
– Suppliers' order number or invoice number
– A summary of your problem

	Address	Tel	Fax/Email
U.S & Canada & Latin America	GE Healthcare 3200N, Grandview Blvd, Waukesha, WI 53188	+1 800 437 1171	
Europe & other Countries	The Competitive Advantage Italy 27100 Pavia Via Sacco 7	+39 382 303990	Roberto@caciolli.191.it
Asia & all other Countries	AIRSYS No.28, LuGu St., Shijingshan Dist. Beijing P.R.China 100040	+86 10 68656161	+86 10 68652453

Table- 2 Customer Service Information

1.3. Package Contents (MEDICOOL 10.0 P6 R407C)

S/N	Item	Quantity	Check & Mark
1	10kW Gradient/Cold head Compressor Water Chiller	1	□
2	Key for control box	2	□
3	Fuse 5X20 250VAC, 0,25A (In the fuse terminal block on chiller)	2	□
4	Label “Gradient Coil ” and “Cryo Cooler ”	2	□
5	User Manual	1	□

Table- 3 Packing list of unit box

S/N	Item	Quantity	Check & Mark
1	19.1mm (3/4”) hose barbs	2	□
2	19.1mm (3/4”) hose clamps	6	□
3	19.1mm(3/4”) male halves of quick disconnect fittings	2	□
4	30.5m (100ft) lengths of 19.1mm (3/4”) flexible rubber hose	2	□
5	1-meter section of 12.7mm (1/2”) ID rubber hose with 12.7mm(1/2”) to 19.1mm(3/4”) hose adapter fitting to allow the 3/4” hose to be connected to the 12.7mm(1/2”) fittings on the cold head compressor.	2	□
6	Remote control panel	1	□
7	Control cable from remote control panel to chiller 30.5m (100ft)	1	□
8	Terminals with size of 1.0mm2(1.55X10-3in2), 1.5mm2(2.33X10-3in2), 6mm2(9.3X10-3in2)	6 for each	□
9 10	50/50 Mixture of propylene glycol and de-ionized water with additive of rust inhibitor and yellow dye	81L (21.5 gal)	□
11	Funnel	1	□
12	Teflon tape 20mX25mmX0.1mm	1	□
13	Special Wrench	1	□

Table- 4 Packing list of spare parts kit

1.4. MEDICOOL 10.0 P6 Brief

The Medicool 10.0 P6 water chiller is a single-loop device, which provide minimum 23.1 Lpm (6.1Gpm) of temperature constant coolant for medical equipment, or any process/machine requiring temperature control.

It is consisted of a refrigeration unit, coolant reservoir and pump, integrated together in a shelter that allows the unit to be operated indoors or outdoors. It has a microprocessor controller, LCD set/read and readout in Celsius Degree (°C). The chiller also provides a remote control panel that allows the user to control the chiller wherever it is located. indoors or outdoors. The remote control panel allows the user to turn the chiller on or off, monitor chiller's temperature, set temperature, check alarm. The refrigeration system is able to provide cooling as needed under precise temperature control, greater temperature stability and long life of compressor. In general, this chiller has been designed for long life and ease to use.

Environmental conditions:

- Non-operating:

Ambient Temperature: -34°C (-29.2°F) ~ +55°C (131°F)

Altitude: 120m (400 ft) below to 3600m (11,808 ft) above sea level

Magnetic Field: 30 Gauss

- Operating:

Ambient Temperature: -30°C (-22°F) ~ +43°C (110°F)

Altitude: 30m (100 ft) below to 3600m (11,808 ft) above sea level

Magnetic Field: 30 Gauss

1.5. Specifications

ITEMS	UNITS	SPECIFICATIONS 50Hz	SPECIFICATIONS 60Hz
Cooling Capacity (1)	kW	10.0	11.6
Refrigerant		R407C	
Refrigerant Charge	kg	5.0	
Power Supply Voltage	Rated	3φ-380/400	3φ-460/480V
	Scale	3φ-342~440V	3φ-414~506V
Total Input Power (peak)	kVA	8.8	10.9
Total Input Power (continuous)	kVA	5.0	6.2
Total Current	A	13	15
Air Heat load when cooling gradient coil	kW(BTU/hr)	15.2(55600)	16.8(61440)
Air Heat load when cooling coldhead compressor	kW(BTU/hr)	13.8(50500)	15.4(56320)
Compressor	**set**	**1**	
Power Input (1)	kW(HP)	3.64 (4.9)	4.30(5.8)
Current (1)	A	6.7	7.9
Maximum Current	A	8.7	8.7
Starting Current	A	51.0	51.0
Axial Fan.	n	1	
Horse Power	HP	0.4	0.53
Current	A	1.25	1.7
Nominal Air Flow	M^3/h	5000	6000
Refrigerant-water Heat Exchanger	**set**	**1**	
Water Flow (1)	L/min(GPM)	23.2(6.1)	23.2(6.1)
Water Resistance (1)	kPa	22.5	22.5
Refrigerant Circuits	**set**	**1**	
Capacity Control Stages	n.	2	
Capacity Control Methods		hot gas bypass	
Water Pump	**set**	**1**	
Head	m	52.0	75.0
Horse Power	HP	1.48	2.01
Current	A	3.0	3.6
Coolant		50/50 mixture of propylene glycol and de-ionized water solution, with additive of rust inhibitor and yellow dye	
Water Tank Content	Liter(gal)	33(8.7)	
Set Point Range	°C (°F)	15°C ~ 25°C (59°F to 77°F)	
Control Accuracy	°C (°F)	±1.0°C (1.8°F)	
Noise	dB(A)	69.0	73.5
Dimension			
Length	mm(inch)	1410(55.34)	
Width	mm(inch)	684.2(26.94)	
Height	mm(inch)	1155(45.47)	
Net Weight	kg(pound)	270(585)	
Weight of Unit when Filled	kg(pound)	310(683)	
(1)-rated conditions: ambient temperature: 43°C (109.4°F); supply coolant temperature: 20°C (68°F);return coolant temperature:25°C (77°F)			

We recommend the main power fuse connected to the chiller should be equipped with 25A。

Table- 4 Specifications

1.6. Diagram of Unit and Remote Control Box

1.6.1. Overall Dimensions of Unit

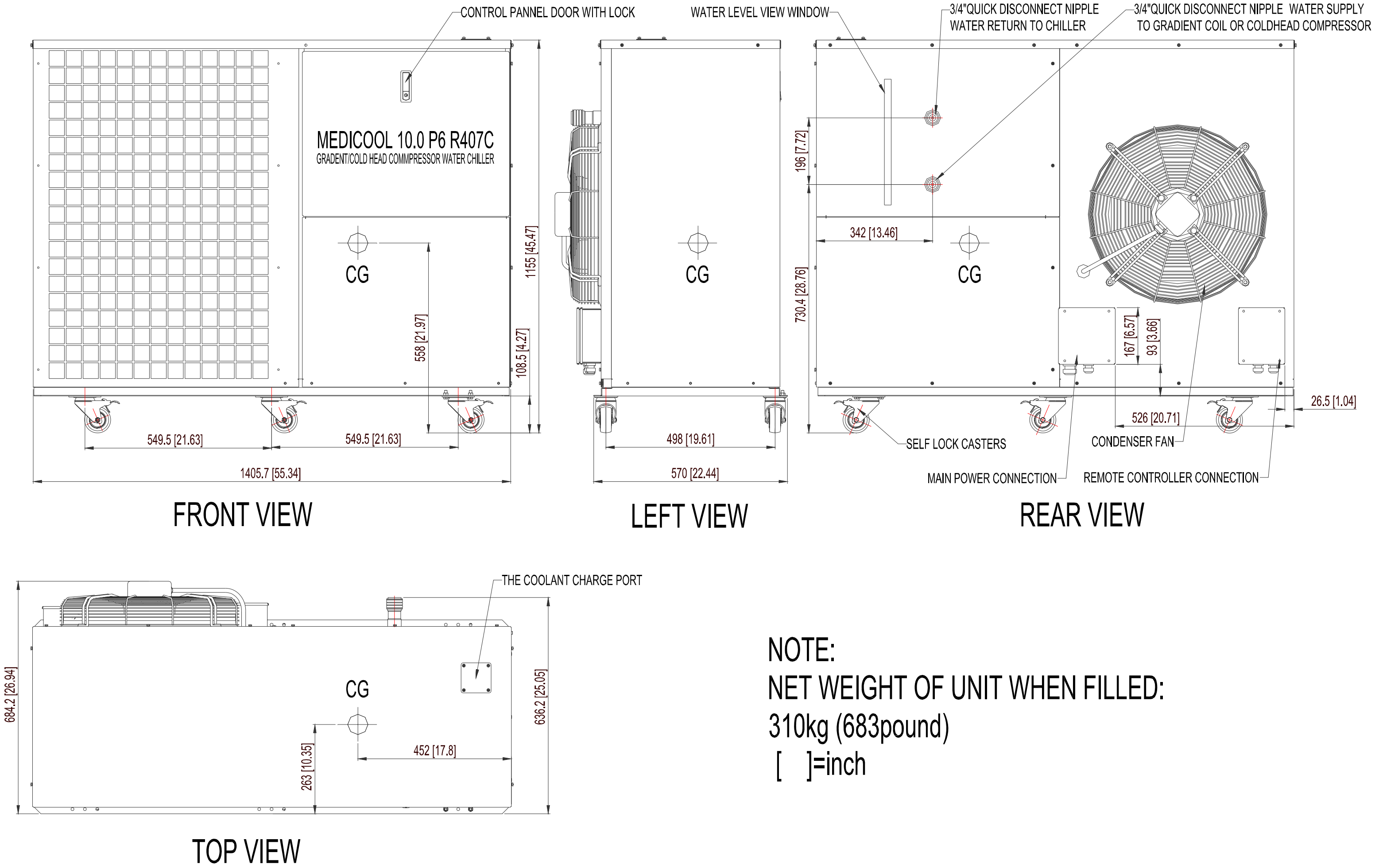


Fig.- 5 Overall dimension

1.6.2. Service Clearances for Single Units

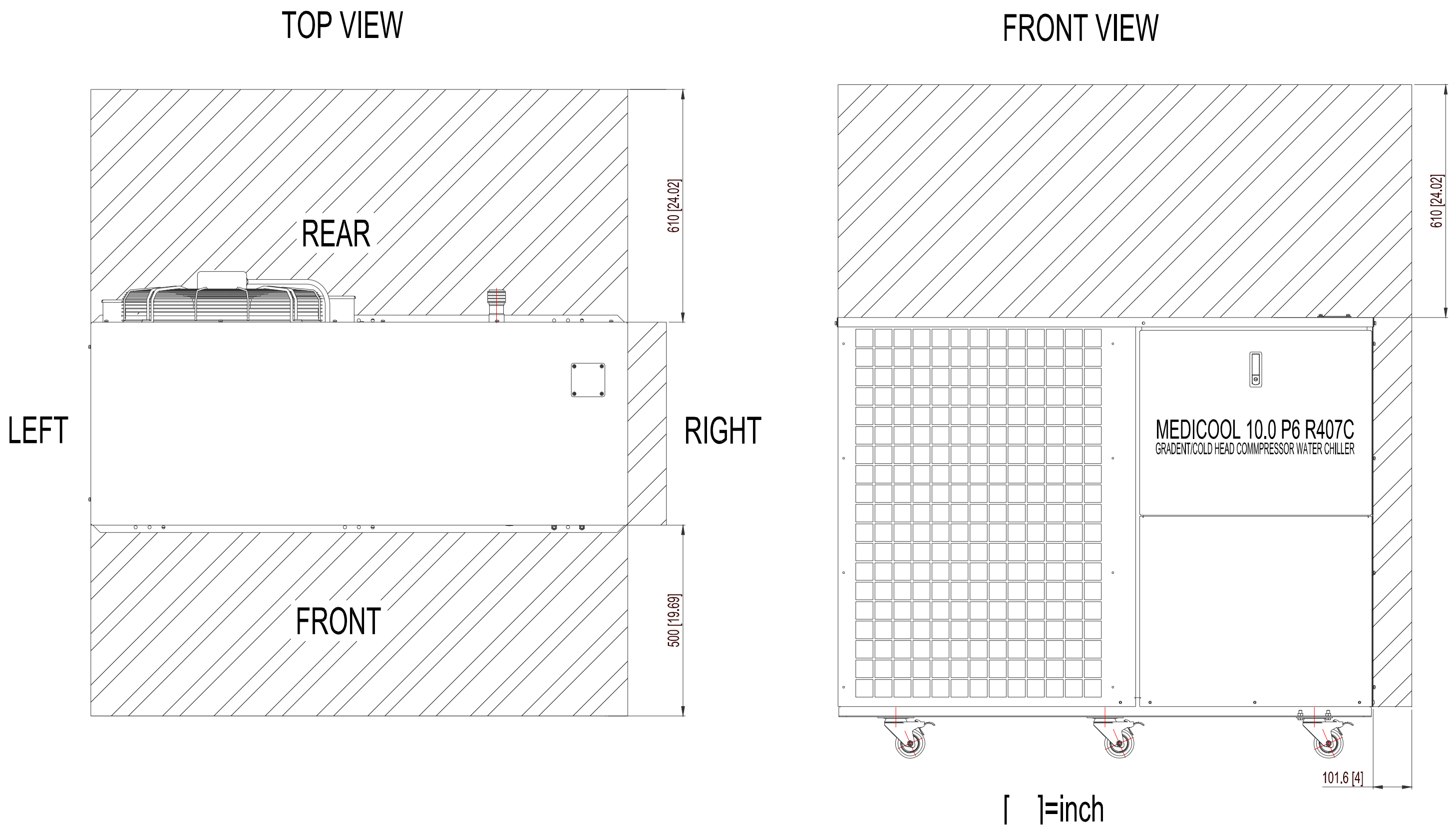


Fig.- 6 Service clearance for single unit

1.6.3. Service Clearances for Two or More Units

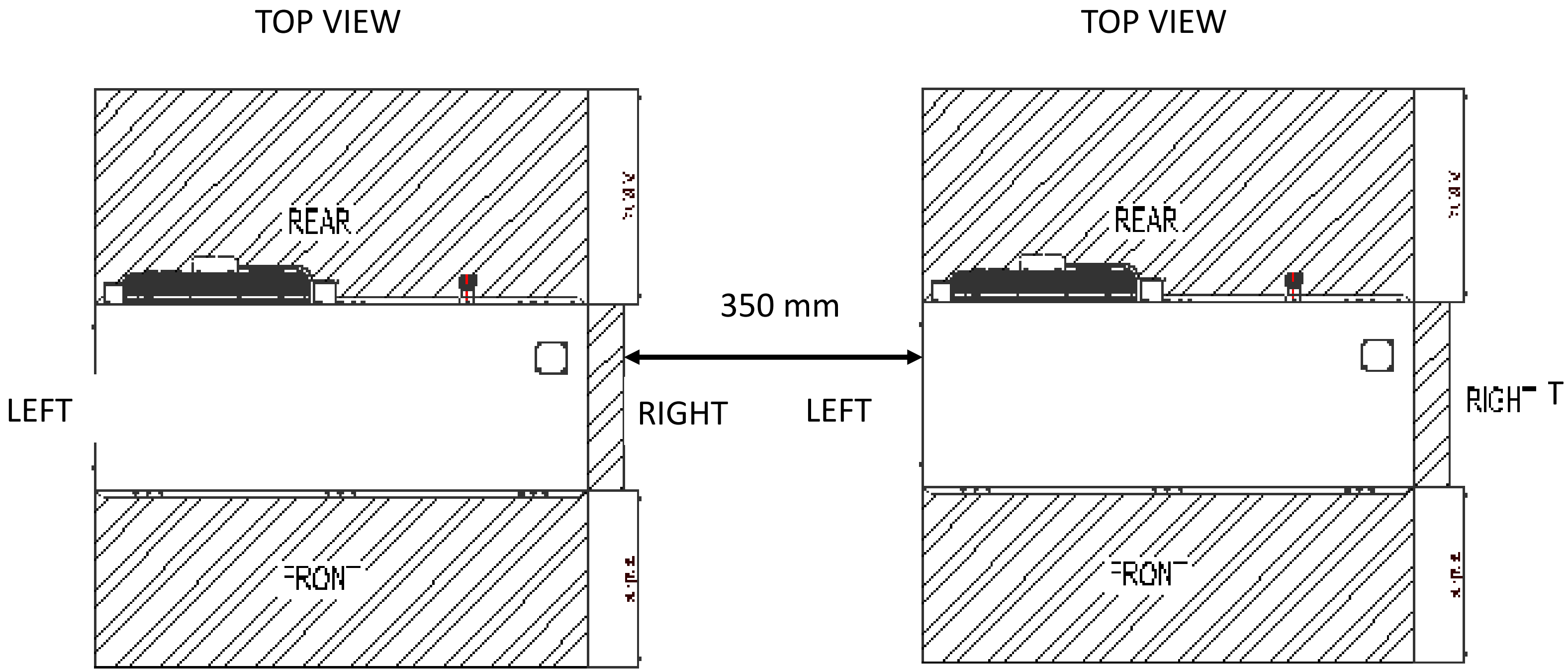


Fig.- 7 (a)Service clearances for two or more units (recommended)

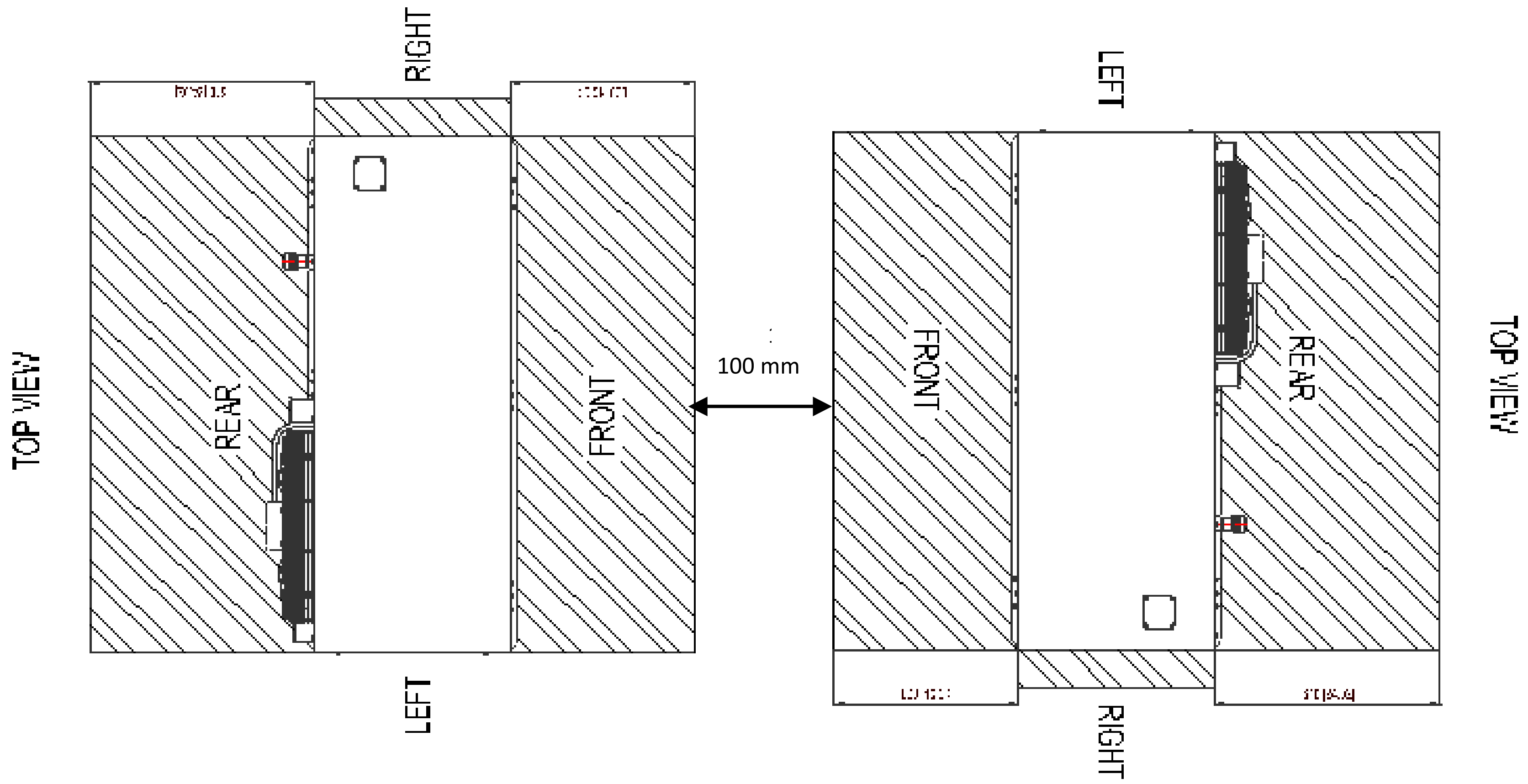
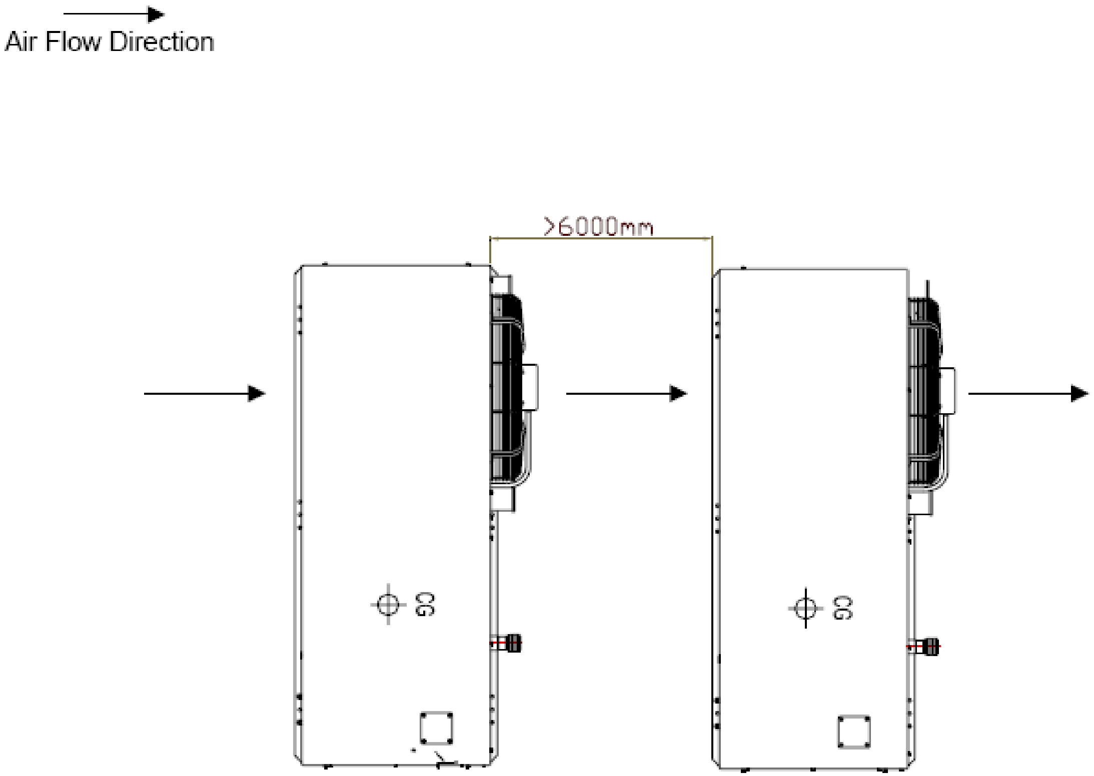
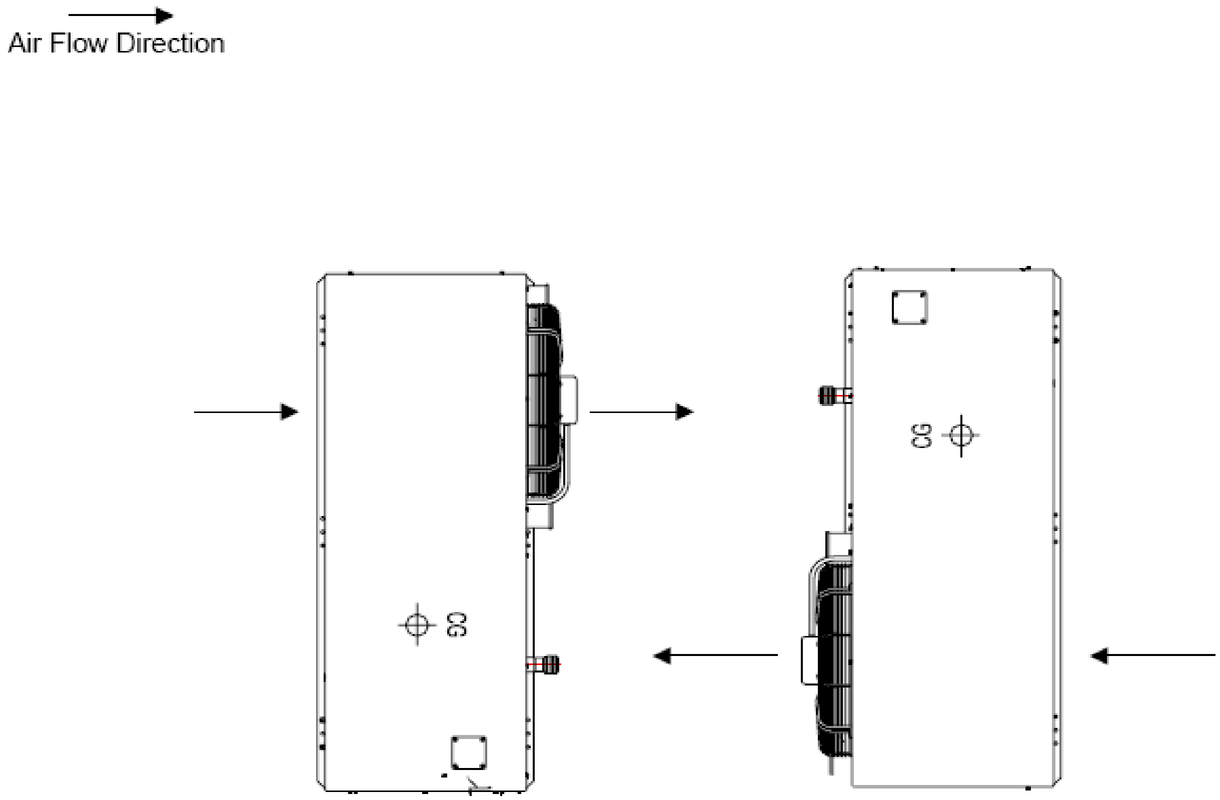


Fig.- 8 (b) Service clearances for two or more units(recommended)



Ensure the distance between chillers MUST great than 6 meters for this layout.

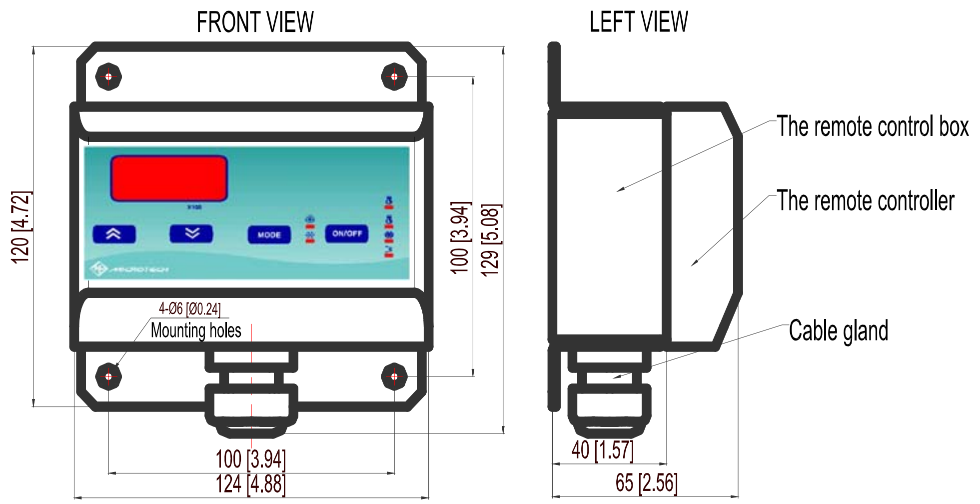
Fig.- 9 (c) Service clearance for two or more units



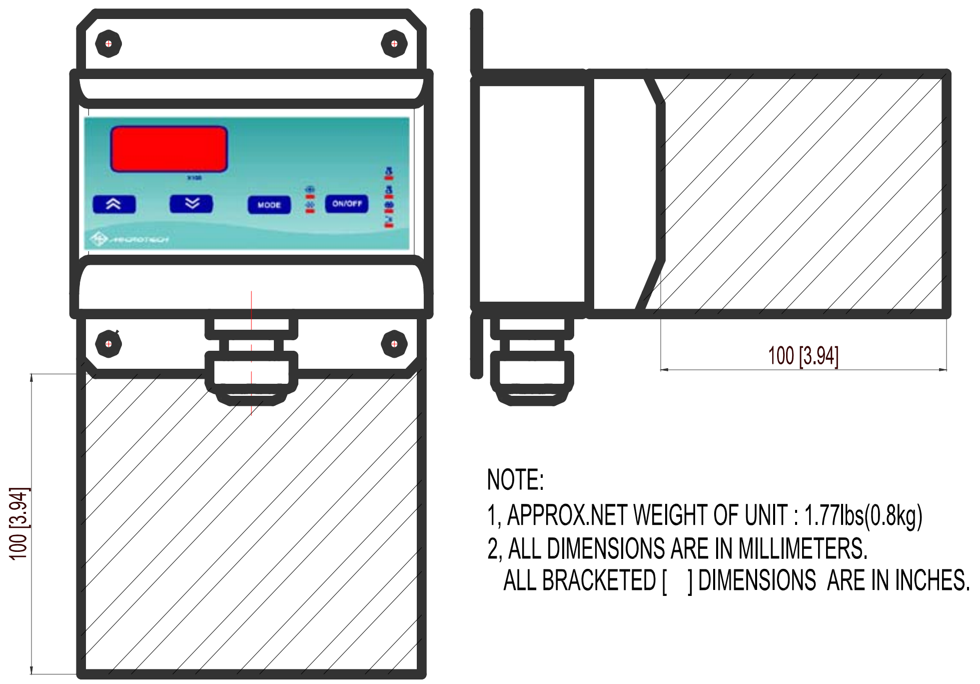
DO NOT let two fans blow directly to each other, which will cause chiller stop for high condenser temperature/pressure alarm.

Fig.- 10 (d) Service clearance for two or more units

1.6.4. The dimension of the remote control box



OVERALL DIMENSION



SERVICE CLEARANCE FOR REMOTE CONTROL BOX

Fig.- 11 Dimension of remote control box

Section 2 – Preparation for Installation

2.1. Installation Responsibility and Installation Tools

Task	Customer	GE Service	Air-Sys	Who pays
Unload Chiller from truck	X			Customer
Move chiller to equipment room or outdoor concrete pad and mount in accordance with local code	X			Customer
Connect customer-supplied power cable from MDP to Chiller	X			Customer
Install water lines to chiller	X			Customer
Install Remote Control Panel in Equipment Room	X			Customer
Fill chiller with Glycol	X			Provided
Start chiller Customer and GE verify proper phase rotation and no leaks	X		X	NA
Perform final Inspection of chiller and verify proper operation			X	Service provider

Table- 6 Installation responsibility

NOTE: It is the customer's responsibility to choose the correct wire size according to line length, insulation type, wire routing and to comply with any local code requirements.

IMPORTANT:
Prior to Installation, you please call (insert proper phone numbers), and provide the following information,

Hospital Name and Address
Serial number of your Chiller
Date of Installation
System ID or FDP of magnet
Local FE name and Phone

You will then receive the code to unlock door to start the chiller.

Tool	Spec	Quantity	Purpose
Line Type Screwdriver	50X2.5	1	Wiring
Line Type Screwdriver	65X5	1	Piping, Wiring
Cross Type Screwdriver	2#	1	Open the junction box
Spanner	300#	2	Piping
Megohm meter	500V	1	Testing

Table- 7 Installation tools

2.2. Installation Requirements

The chiller can be installed indoors, outdoors or outdoors on rooftop according to your field condition. You can mount the chiller follow the manual. Any other mounting functions should be discussed with the AIRSYS or field engineers.

2.2.1. Height of the Location

The distance between the MEDICOOL chiller and gradient coil/shield cooler must be less than 30 m (98ft). The height difference between Medicool chiller and gradient coil must be less than 3 m (9.8ft). The power distribution panel should be placed as closer as it could to the chiller.

2.2.2. Air Considerations

The air inlet and outlet are on the front and rear sides respectively. Airflow blows in and out of the unit will affect performance. The minimum clearance of machine installation is required, as Fig. 6.1, to ensure adequate airflow.

When two or more units are closely installed, the minimum clearance for each machine must be met, as Fig.- 7, 8, 9, 10.

Avoid the axial fan directly face to monsoon, it can cause high pressure alarm. Insufficient ventilation could gradually increase the condensing temperature and cause the intervention of pressure switch. The unit installed outdoors in extremely ventilated area may be affected by the seasonal wind. In such case, it is wise to install anti-wind shields by customer.

2.2.3. Indoor Use

Move the chiller to a strong, level surface (1 cm over 300cm), and fix the casters.

2.2.4. Outdoor Use on Concrete Ground

Remove the casters and using the six middle holes to firmly fix the chiller on a concrete ground. (See Fig.- 12)

Concrete ground requirement:

Concrete ground used for mounting the unit should be a level surface, which is 1/300 cm max allowed and be properly supported to prevent sedimentation. A concrete made area of 50.0cm (59 in) x66.0cm (26 in) at strength of 17.23MPa (2500 psi) min (4 inches thickness recommended) is needed to place the chiller. The concrete footing should meet or exceed the local code requirements.

2.2.5. Outdoor Use on Rooftop

Remove the casters and using the six fixing holes to firmly fix the chiller on a level surface on rooftop, which is 1/300 cm max allowed.

Fig.- 13 shows the mounting holes location.

Note: In case of extremely bad weathers, like snowstorm, snow and ice may influence the proper working of the fan. So it is recommended to shelter unit to protect the fan against the bad weathers.

Note: Be careful of the height of location place to ensure no circumfluence of coolant to the system.

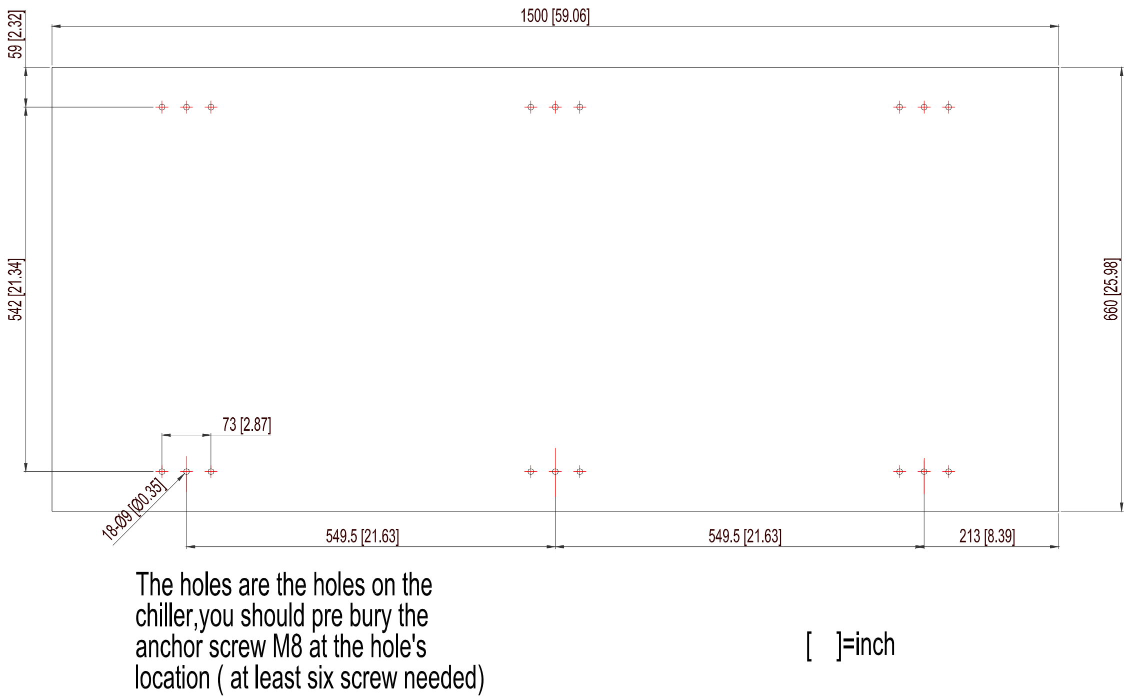


Fig.- 12 Concrete foundation

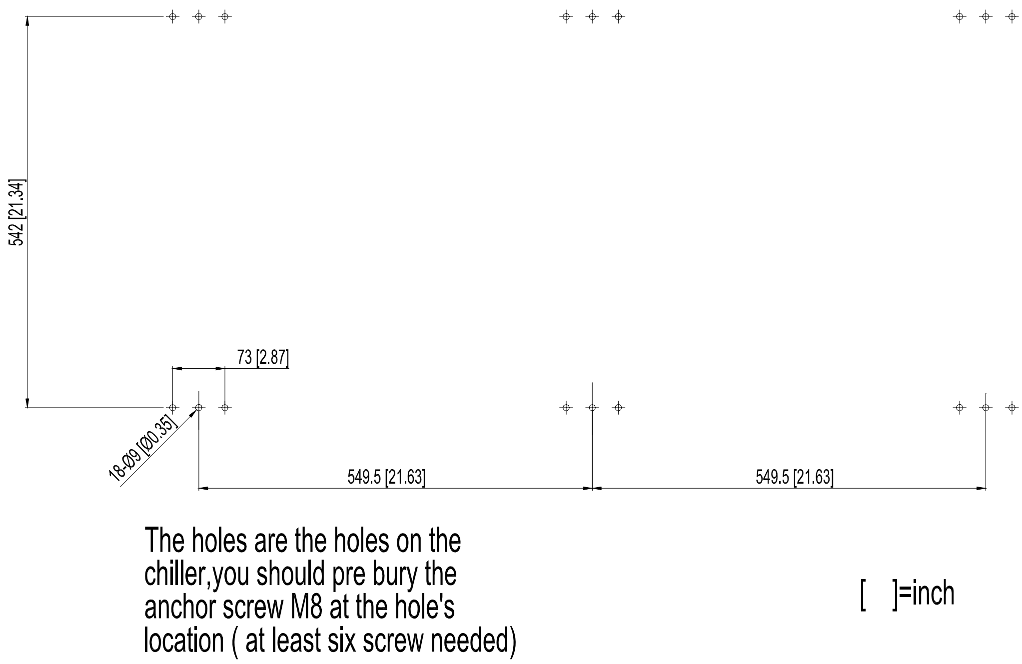


Fig.- 13 Mounting holes location

2.3. Installation Configuration

.2.3.1. Power Connections

2.3.1.1. Power Requirements

ITEM	REQUIREMENT
Wire (customer supplied)	The maximum wire size is 6 AWG (16 mm^2), the minimum wire size is as needed to meet the voltage specifications of Table 4
Power Consumption	6.2KVA (maximum, continuous)
Voltage Requirements	380/400 VAC@ 50 Hz 460/480 VAC@ 60 Hz
Circuit Size	15AMP(minimum)

Table- 8 Power requirements

Note: The unit uses a front door lock that prevents the customer from opening the front door without switch to the "off" position.

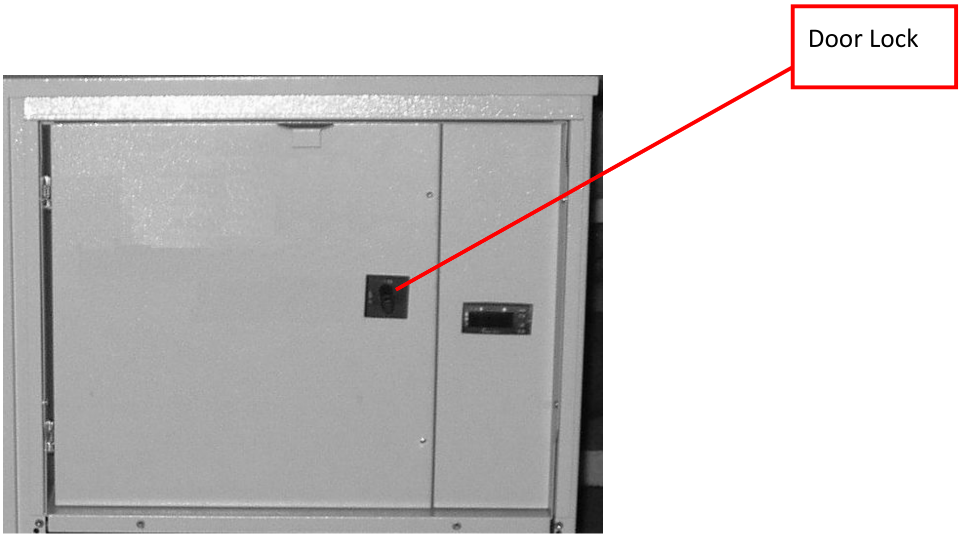


Fig.- 14 Customer lockout

2.3.1.2. Electrical Connection

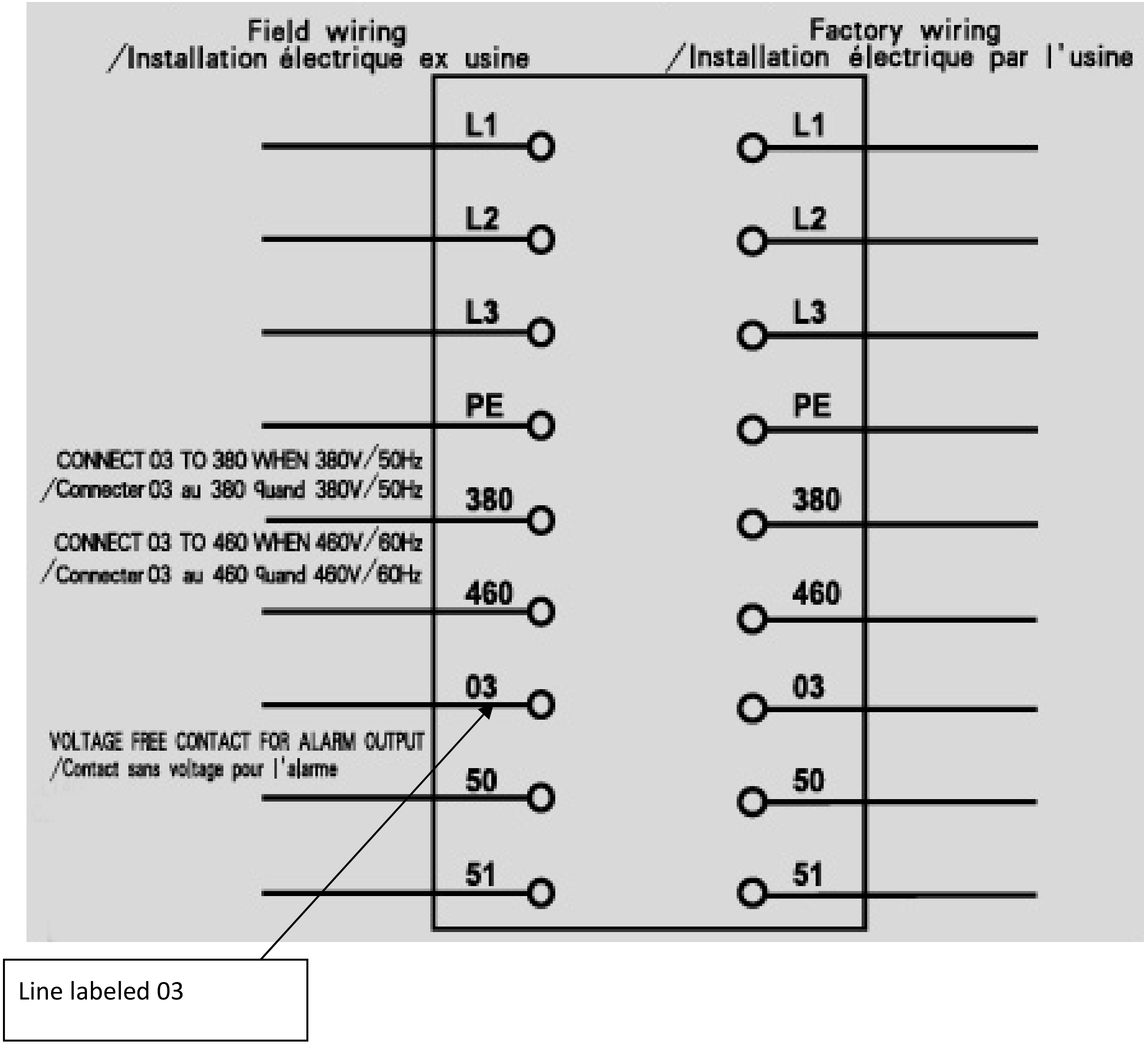


Fig.- 15 Main power connection

The left column (with nine binding posts) is for power wiring. Put the power line (four wires) through power inlet port of the box. Get 3 phase wires connected to the right three binding posts labeled L1, L2, L3 and ground wire to PE post.

If the power supply is **380/400 VAC 50 Hz**, connect the line labeled 03 (see the Fig.- 15) with the opposite post of 380.

If the power supply is **460/480 VAC 60 Hz**, connect the line labeled 03 with the opposite post of 460.

Note: If the Chiller will be connected to 50 Hz power, refer to section 2.3.1.3, and section 3.4, on how to properly configure H43 parameter of the chiller for 50Hz operation.

The binding posts labeled 50 and 51 are used to connect the external alarm indicator (ring bell or flash lamp) supplied and powered by user.

The right junction box is for remote control panel connection. Connect the line labeled 24, 25, 26 with the binding post-labeled 24,25,26 respectively. In this box, there are two binding posts labeled 10,11, they are used for remote on-off function.

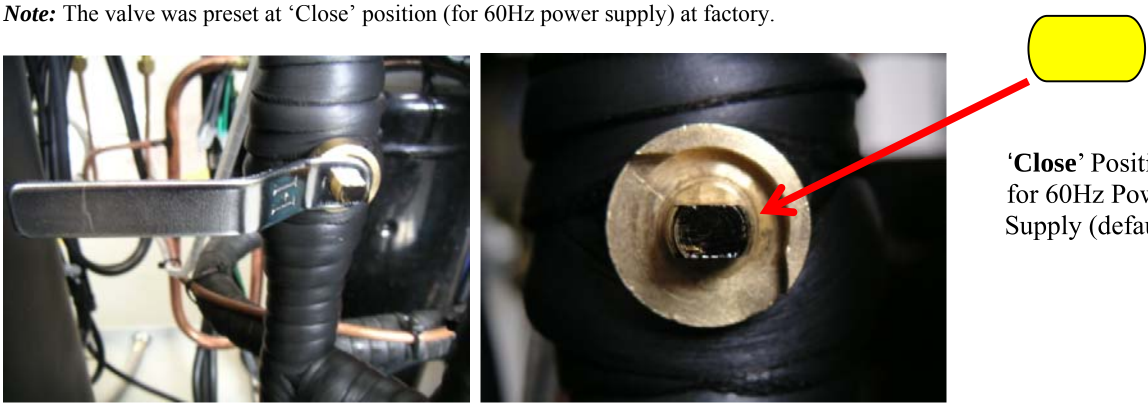
2.3.1.3. 50Hz/60Hz Configuration

Note: Be aware of wire jump configuration per section 2.3.1.2.

Step 1: If the power supply is 50 Hz, change parameter H43 (see appendix A) to 0. If the power supply is 60 Hz, change parameter of H43 to 1. Please refer to section 3.4 to get know how to change the parameter.

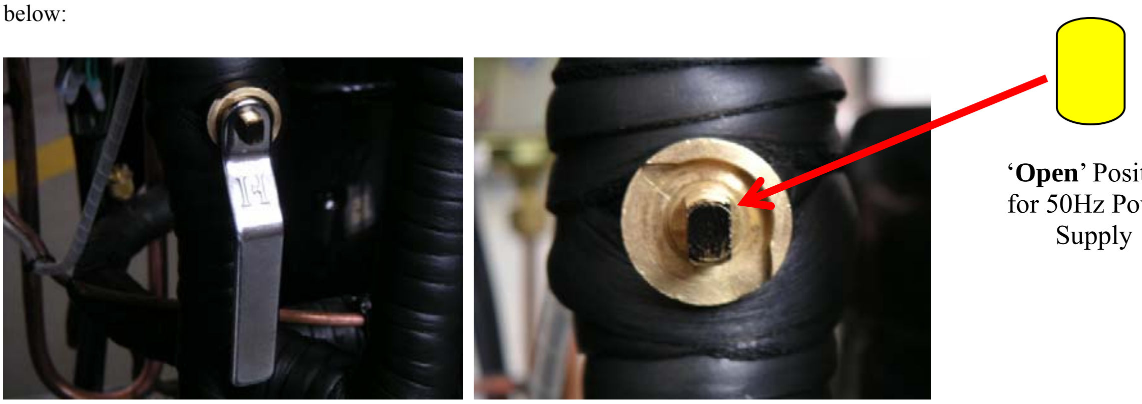
Step 2: Remove screws to open the rear lower panel. Adjust the valve without bonnet (shown in Fig 18) by the special wrench in accessory kit to 'open' position for 50Hz power supply in site.

Note: The valve was preset at 'Close' position (for 60Hz power supply) at factory.



'**Close**' Position for 60Hz Power Supply (default)

If in ***50Hz*** power supply, please ensure the valve switch to ***OPEN*** position by means of the special wrench as shown below:



'**Open**' Position for 50Hz Power Supply

Fig.- 16 Configuration of the valve for different power frequency

After setting the valve in the correct position, Please remove the wrench from the valve and put it into a place away from the chiller (maybe with the user manual), in order to avoid miss operation. Mounting the rear panel after adjustment.

2.3.2. Configuration for Chillers Used in –35ºC Ambient Temperature

For the low ambient temperature, you should add a relay to bypass the low pressure switch during compressor off time to insure the chiller to work well.

Instruction for installation of the bypass relay:

1. Install the by pass relay aside the KM1 contactor. Please slot the relay into the port on the side of the contactor (the photo shows the wrong status before slotting in). Be sure of that the relay has interlocked with the KM1 contactor automatically. Please refer to Fig.-18, How to install relay. Then connect wire #12 and #13 from the "31X NC" and "32X NC" ports of the relay to the relevant place on the terminal block according to Appendix B: Wiring Diagram

2. If the bypass relay had fixed in factory, the wire #12 should be connected to "43NO" (terminal number on the relay). If you want to start the chiller under low temperature please only reconnect the wire #12 from"43NO"to "31NC".

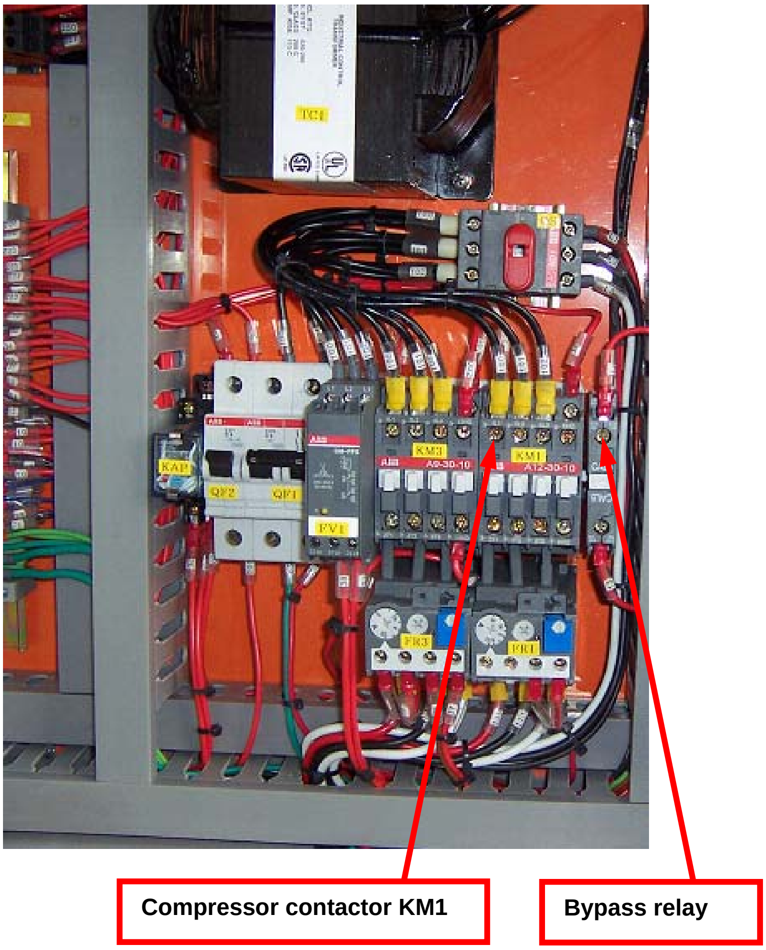


Fig.- 17 Bypass relay

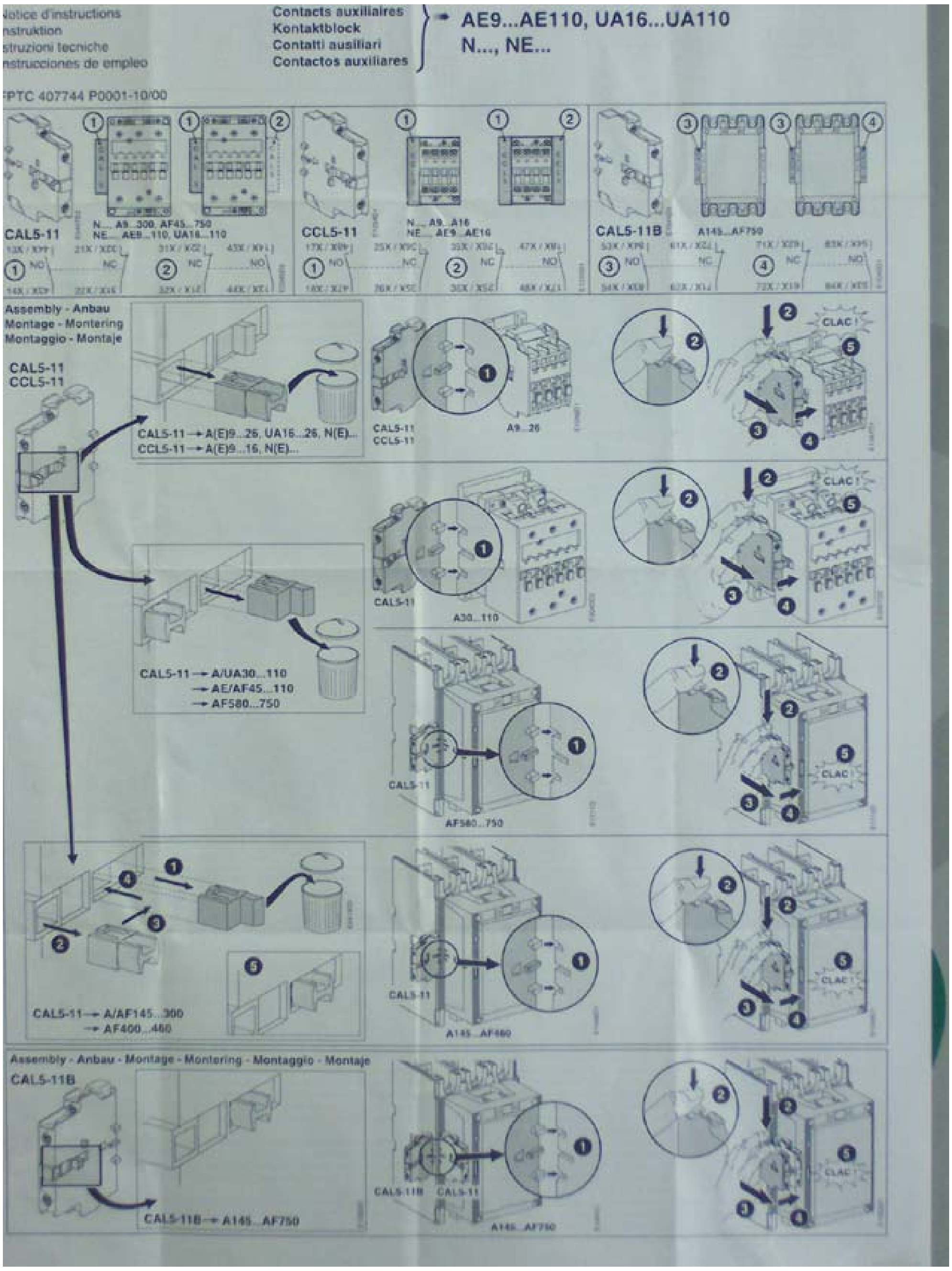


Fig.- 18 How to install relay

2.3.3. Pipe Connections

The remote control panel (with control cable), hose barbs, quick disconnections, hose adapters and hose that may be used during installation are supplied with the chiller. They can be found in the spare parts kit. The copper pipe that may be used in the installation is to be provided by the customer.

2.3.3.1. Indoor Connection

1. Before connecting the water hose with the chiller, check and clean the water hose thoroughly to make sure there is no metal residence or dirt inside. Check the quick disconnects to find any water leaks.

2. Apply the "Gradient Coil " or "Cryo Cooler " label to the front door of the chiller depending on its intended use. These labels can be found in the shipping box.

Connect your closed cooling system to the chiller with hoses or pipes as drawing. The direction of the flow through the system can be controlled by the way the hoses or pipes are connected to the chiller. The "COOLANT RETURN" is drawing liquid into the chiller; the "COOLANT SUPPLY" is pumping cooled liquid out.

Note: The hose connection with the unit should be adjusted smoothly to avoid additional strain that may result in a break or crack to the connector inside the chiller.

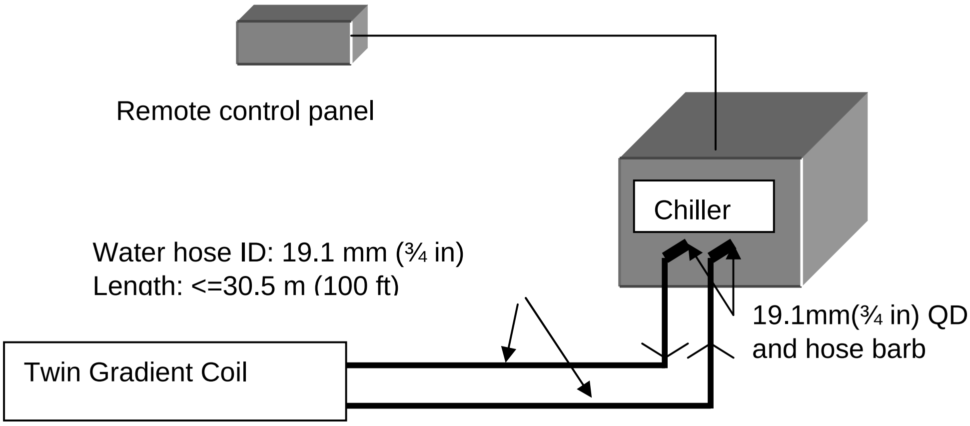


Fig.- 19 Indoor connection with twin gradient coil

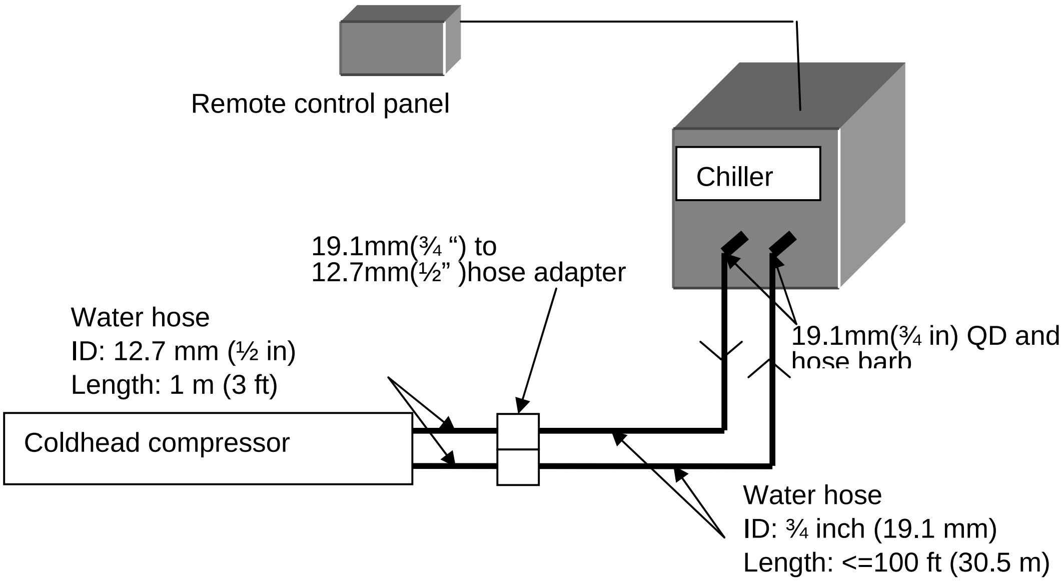


Fig.- 20 Indoor connection with coldhead compressor

The coolant fluid inlet (COOLANT RETURN) and outlet (COOLANT SUPPLY) connections are located on the rear side. They are 19.1mm(3/4") corrosion-resistant, metallic, double-shutoff type quick disconnect together with 19.1 mm (3/4") hose barb and hose clamp. Please refer to figure below (Fig.-21).

Fig.- 21 Quick disconnection

2.3.3.2. Outdoor Connection

Note: At outdoor installations, the copper pipe must be thermally insulated.

The copper pipe is connected to the chiller via a short section of rubber hose. It is user's responsibility to supply ¾ inches hose barbs and clamps for connecting the hose to the copper pipe. If hosepipe is used outdoor, thermally insulation is needed on whole pipe to prolong its life.

Please refer to Fig.- 22 and Fig.- 23 for connecting of Gradient Coil or Coldhead Compressor with chiller.

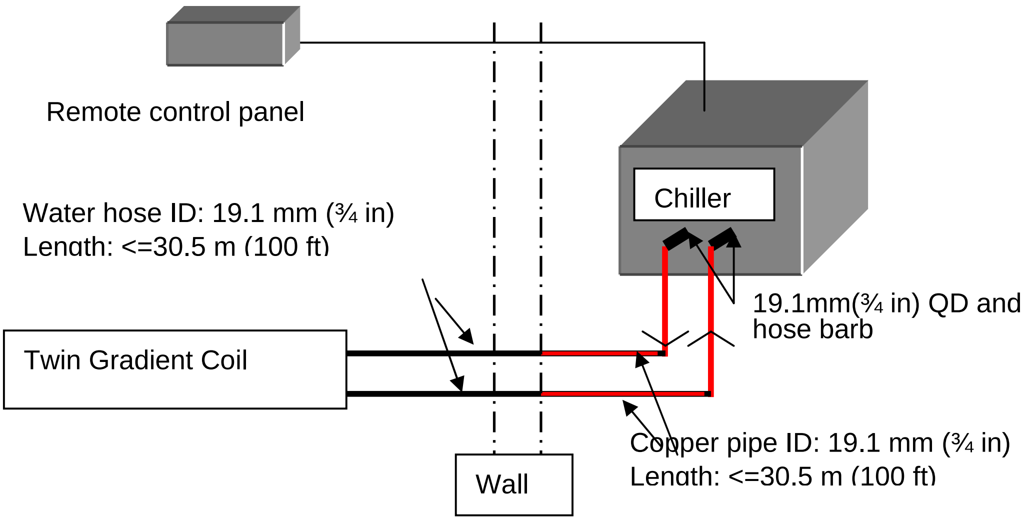


Fig.- 22 Outdoor connection with twin gradient coil

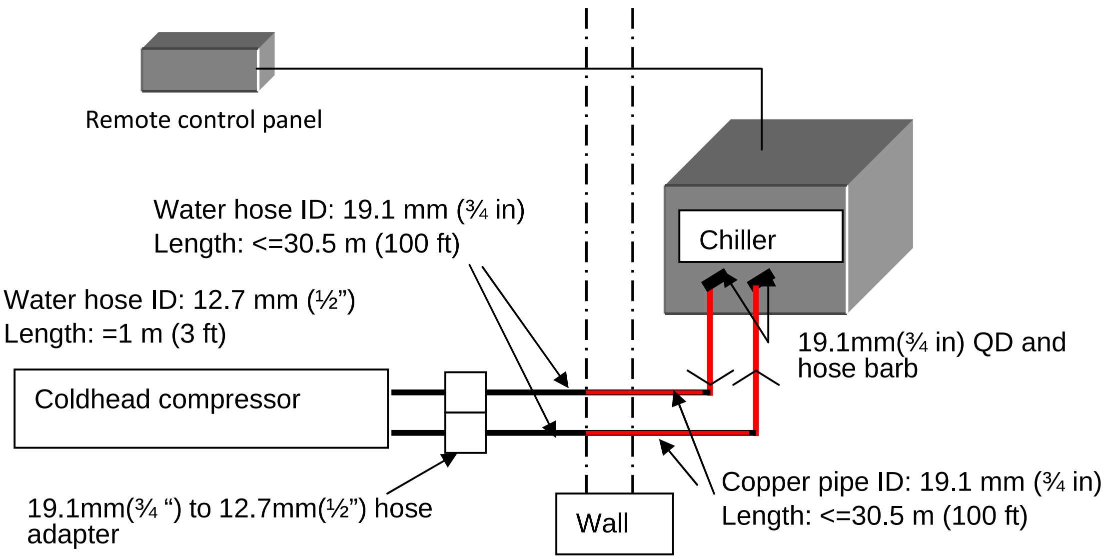


Fig.- 23 Outdoor connection with coldhead compressor

Section 3 – Operation

3.1. Introduction of the Control Panel & Remote Control Panel

The MEDICOOL is operated via the control panel on the unit and/or the remote control panel.

The control panel on unit has a LCD display and two input keys: 

, 

The function of the keys are as following:

On-off Reset alarms 	Reset alarms, and turn the chiller on and off. Press once to reset all manually reset alarms not currently active. Press down the key and hold for 2 seconds to turn the unit from on to off or vice versa. When chill switch off, only the decimal point remains on the display. In menu mode this key acts as a SCROLL DOWN or VALUE DECREASE key.
Set Running Mode 	In first menu, display the supply water temperature. By press this key to change the running mode of the unit between cooling and stand-by. In menu mode this key acts as a SCROLL UP or VALUE INCREASE key.
Mode/On-off key Combination 	Pressing the “mode” and “on-off” keys at the same time. If you press both keys at the same time and release within 2 seconds, you will move one level deeper in the display menu. If you press both keys for more than 2 seconds you will move one level up. If you are currently viewing the lowest level in the menu and you press both keys and release within 2 seconds, you will go up one level.
	Normal *display* shows: ● Regulation temperature of degrees Celsius with one decimal, or in degrees Fahrenheit without decimal. ● The alarm code. If multiple alarms are active, the one with priority will be displayed, according to the Table of Alarms. ● When in menu mode, the display depends on the current position; labels and codes are used to help the user identify the current function. ● Decimal point: when displaying hours of operation, indicates that the value must be multiplied x 100

Table- 9 Function of control panel keys

LED Indicator

1	Compressor led ➢ ON if compressor is active ➢ OFF if compressor is off ➢ BLINK if safety timing is in progress
2	Capacity step led ➢ ON if capacity full load ➢ OFF if capacity half load ➢ BLINK if safety timing is in progress
	No use for MEDICOOL 10.0 P6 R407C
	No use for MEDICOOL 10.0 P6 R407C
	No use for MEDICOOL 10.0 P6 R407C
	Cooling led ON if the controller is in cooling mode

Note: If neither the HEATING LED nor the COOLING LED is on, the controller is in STAND-BY mode

Table- 10 Display of LED

The remote control panel display is an exact copy of the instrument with same LEDs

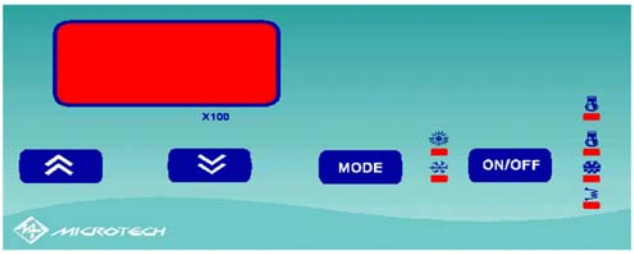


Fig.- 24 Remote control panel keyboard

Remote control panel has exactly the same functions as described in the display section. The only difference is that the UP and DOWN keys (to increase and decrease value) are separated from the 'MODE' and 'ON/OFF' keys.

3.2. Preparation for First Start

Note: Before first start, please double check electrical control system include control panel, all junction box inside or outside chiller to find and replace burned or frayed wiring, and check insulation, fix loose connections.

3.2.1. Coolant Fill

Fill the chiller with coolant Mixture per following steps:

Note: Important! Only use the coolant ship with the chiller. If additional coolant is needed, please order from your service supplier, part number 2297672, one gallon of properly-mixed coolant.

Note: The provided glycol is 50/50 mixture with additive of rust inhibitor and dye in yellow.

Note: Check coolant system to ensure it is clean enough before chiller start-up. Otherwise flushing work should be conducted first.

Screw off top cover, put funnel in the coolant charge port. Fill the reservoir with fluid. When full, remove the funnel but do not place the cap at this time. Check hoses and fittings for tightness and be sure no bends or crimps on the hoses.

Note: After fill tank with coolant, follow instruction attached on front door to discharge air in pump. It is very important ot protect pump seal from damaging in short period of running.

Start up the chiller per section 3.3.1.

Press the on/off button, then mode button, to turn chiller on (refer to section 3.3, Normal Start and Stop the Chiller). The chiller will begin pumping liquid through the system at once. Meantime, the fluid in the reservoir will decrease as part of it fill into the system. Add coolant until the level in the reservoir stops going down. This means that your system is filled and the air has been purged out completely. Continue to fill the reservoir until level reaches "**COOLANT CHARGE LEVEL**" line. Replace the top cover before normal operating.

3.2.2. Outlet Fluid Temperature Setting

Turn on the controller (refer to 3.2), the actual outlet fluid temperature is displayed.

Required operating temperature can now be set. Press the "mode" and "on-off" keys at the same time and then release within 2 seconds for three times, the set cooling point is displayed. Press the  or  key to change the value (°C) to the target point. Press the "mode" and "on-off" keys at the same time more than 2 seconds and then release for three times, return to stand-by.

Note: Outlet temperature set point should be lower 1~1.5°C than you wanted. For example, *the chiller has been preset on 19°c (66.2°f), therefore the stabilized outlet temperature is 20°c (68°f).*

3.3. Normal Start and Stop the Chiller

Caution! Be sure that the power supply is same as power supply specification on the label.

3.3.1. Start the Chiller

Open the front door and turn on the door lock power switch (Fig.- 14). The display will show a decimal point or other status.

NOTE: For the first start up, to heat the lubricating oil of the compressor, you should wait for at least 2 hours after turning on the unit.

Press the “ON-OFF” key on the control panel or remote keyboard, hold for 2 seconds to turn on the controller. The display shows the outlet fluid temperature.

NOTE: Wrong power phase setting will not start up the machine without error code displayed. Change the sequence of any two of L1, L2, L3 in the power junction box to correct the mistake and let the chiller power on.

Press the “MODE” key to turn on the chiller. The pump is active first, then the compressor will start after coolant temperature rise above 20°C (68°F).

NOTE: When the power restored after power failure, the controller will be back to the status before the power failure.

3.3.2. Stop the Chiller

Press the “MODE” key on the control panel or remote keyboard to stop the chiller.

Press the “ON-OFF” key on the control panel or remote keyboard for 2 seconds to turn off the controller.

3.3.3 Start-up Check List

It is service engineer’s responsibility to fulfill the Start-up Check List, in Table-10. The list should signed by GE Healthcare field engineer and/or end-user as appproval.

Start up Checklist of Medicool 10kW Chiller

Model:Medicool 10.0 P6 R407C   Airsys Refrigeration Engineering Technology Co. Ltd.

Customer: *Document NO.* *GEService001*   *Revision:2.1*

Location: *Unit Serial NO.:*

Service company:

Service Engineer signature: *Date:*

	Check list	Initial status	Correct action	Status after	Remarks	Requirement
1	Check the location of the unit, it should be on a strong, and level surface. The casters should be locked for an indoor installation, tied down for an outdoor installation. Chiller can be located 30 m above or 3 m below the gradient coil or cryogen compressor. See Section 2.2 in the user manual.					-3m<H<30m, lean < 1/300
2	Check coolant piping for loose connections and leaks.					No leak
3	Leak test refrigerant system.					No leak
4	Check the coolant in the reservoir, add coolant if it is below the charge level.					Fill to "Coolant Charge Level Line"
5	Check electrical wiring, including control panel and all junction boxes, for loose connections or misplaced wiring, especially the main power wiring line "03". ("03" is used to switch between 50Hz and 60 Hz applications) See Section 2.4 in the user manual.					
6	Check the voltage, and phase rotation. Verify voltage setting of chiller to match power supply in site. See Section 2.4.1 in the installation manual.					Voltage variable range should be within +/-10%
7	Verify the pump outlet valve is adjusted to the correct outlet pressure after start up. Adjust pressure to obtain correct flow of 6.1 GPM. Do not allow the pressure to exceed 74 Psig.					0.45MPa+0.5MPa/6.1 GPM
8	Verify the clearances around the chiller meet the requirements in Sections 1.5.2 in the installation manual.					Front>500mm, Rear>600mm
9	Verify the proper operation of the pump system, record the following running data;					
9-1	Flowrate					min. 3GPM/12.5 LPM
9-2	Outlet pressure					<0.52MPa
9.3	Running current of each phase	L1:		L1:		11.0<A<17.0
		L2:		L2:		11.0<A<17.0
		L3:		L3:		11.0<A<17.0
9.4	Running voltage of each phase	L1&L2:		L1&L2:		(380/460)+/- 10%
		L1&L3:		L1&L3:		(380/460)+/- 10%
		L2&L3:		L2&L3:		(380/460)+/- 10%
10	Verifying proper operation of refrigerant system, record the following operating data .					
10.1	Compressor discharge pressure					<2400kpa
10.2	Temp of compressor inlet					>0℃
10.3	Coil temperature (read from PLC)					<55 deg ℃/ 131 deg ℉
10.4	Condenser air inlet temperature					<45℃
10.5	Supply coolant temperature (read from PLC)					18.0℃<T<20.0℃, +/- 1℃
10.6	Return coolant temperature (Read from PLC)					24.5℃<T<28.5℃
10.7	Compressor suction pressure					>436kpa
10.8	Compressor input voltage of each phase	L1&L2:		L1&L2:		(380/460)+/- 10%
		L1&L3:		L1&L3:		(380/460)+/- 10%
		L2&L3:		L2&L3:		(380/460)+/- 10%
10.9	Coil temperature (read from PLC)					<55 deg ℃/ 131 deg ℉
10.10	Compressor discharge pipe temperature at 4 to 8 inches from the discharge port.					<90 deg ℃/194 deg ℉
10.11	Compressor running current of each phase	L1:		L1:		5.0<A<10.0
		L2:		L2:		5.0<A<10.0
		L3:		L3:		5.0<A<10.0
10.12	Chiller total running current of each phase	L1:		L1:		11.0<A<17.0
		L2:		L2:		11.0<A<17.0
		L3:		L3:		11.0<A<17.0
10.13	Check refrigerant sight glass status (add refrigerant and do leak-test if it indicates a low refrigerant charge)					Refrigerant should be dry and no or only few bubble
11	Do a detailed visual inspection of the equipment and a general cleaning. Record any damages.					
12	Check PLC parameters setting by comparing with the table of parameters in manual, and correct the wrong settings.					
13	Verify that the cooling capacity meets the requirement of the GE equipment. Record the supply water temperature every 5 minutes for 30 minutes.			T1: T2: T3: T4: T5: T6:		19 ℃ +/- 1℃

Table- 11 Start-up List

3.4. Parameter Programming

The controller is a fully parameter configurable device. All the parameters have been preset in the factory. The attached parameters sheet (Appendix A, Table of parameters) lists the default parameters setting.

The only parameter, which may be changed during normal operation, is H43. If a parameter is changed, restore it to the default setting as shown in App. A

Parameter programming – Menu levels

Device parameters may be modified using the keyboard. It is arranged in levels which may be accessed by pressing the “mode” and “on-off” keys at the same time.

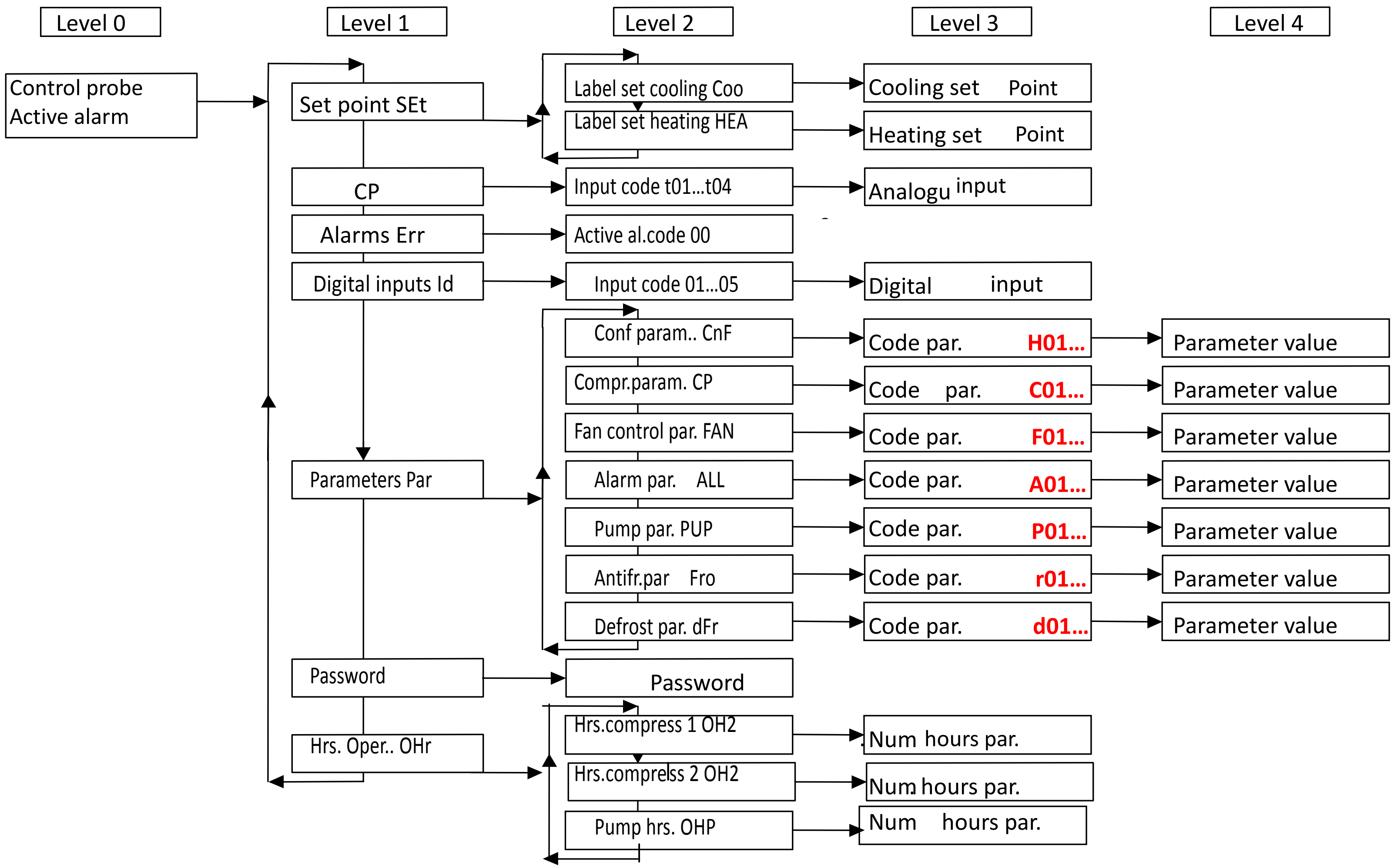
Caution! Be sure that the power supply is the same voltage as specified on the label.

Refer to section 3.1. and Table 8 & 9 to get know how to use the keys and how to move between the menu levels.

Each menu level is identified by a mnemonic code, which appears on the display. The structure is shown as in Table.11.

3.5. Draining the Unit

Turn off the unit, and follow all applicable Lock-out Tag-out procedures. Screw off the front panel, then open the drain valve allowing unit to drain. A proper container is needed to keep liquid out.



NOTE: **Not all the parameters can be seen only if the professional enter in right password.**

Table- 12 Menu structure

Section 4 – Maintenance

4.1. Warranty Policy

The warranty period for the Medicool chiller is ONE year after installation for part(s).

4.2. Preventive Maintenance

Check the unit following steps as below. Fulfill the PM list in Table.12.

1, Check unit performance, parameter setting, and record.

2, Check voltage, current and resistance of compressor, fan and pump.

3, Check contactor of compressor, fan and pump. Replace the contactor if necessary.

4, Check fix bolts of compressor, fan and pump. Tighten the bolts if necessary.

5, Check level of refrigerant through sight glass. Reload if necessary.

6, Check leakage of the refrigeration system. Repair if find any leakage.

7, Check pump seal and coolant circulation to find any leakage at connection or pipe. Repair or replace if find any leakage.

8, Check High/Low Protective Gauge. Replace if necessary.

9, Check system operation pressure (high pressure and low pressure) and operation current.

10, Check temperatures at air inlet and air outlet. Find root cause if the temperature is out of normal range.

11, Check if the machine has abnormal noisy during running.

12, Clean the condenser and cover of the machine. Remove any contaminant on condenser.

13, Check if the cables and pipes are well fixed. Tighten if necessary.

14, Visual check the whole system. Find any potential problems and resolve.

15, Check protection of outdoor unit. (protect from frozen in winter and high pressure in summer)

4.3. Preventive Maintenance Check List

It is service engineer's responsibility to fulfill the Preventive Maintenance Check List, in Table-10. The list should signed by GE Healthcare field engineer and/or end-user as appproval.

Preventative Maintenance Verification Checklist of Medicool 10kW Chiller

Model: Medicool 10.0 P6 R407C

Airsys Refrigeration Engineering Technology Co. Ltd.

Customer:

Location:

Service company:

Service Engineer signature:

Document NO. GEService001 Revision: 1.3

Unit Serial NO.:

Date

	Check list	Initial status	Maintaince action	Status after action	Other comments	Requirement
1	Check the controller for any error codes and investigate as appropriate.					
2	Check the coolant water filter if the flow rate is under the normal value(6.1 GPM / 23.2 LPM) (A pump outlet pressure that exceeds 500Kpa may indicate that the flow rate is under the normal value)					-3m<H<30m, lean < 1/300
3	Leak test for refrigerant system and check for damage in piping.					
4	Check coolant piping inside the unit cabinet for obvious leaks and for worn areas. Report any leaks or worn areas on piping outside the unit cabinet to GE.					
5	Fill the reservoir if the coolant level is below the label "COOLANT CHARGE LEVEL",which is located on the back upper panel.					
6	Check electrical control system including control panel and all junction boxes inside or outside the chiller for burned or frayed wiring and loose connections ,and do the insulation test of the field installing power input line with megohm meter.Record the insulation data.					
7	Check for weak or defective contactors, burned or pitted contacts. Fix or replace contactor if it makes any unusual noise or is the source of high current draw.					
8	Verify the proper operation of the pump system, record the following running data;					
8-1	Flowrate (if available)					min. 3GPM/12.5 LPM
8-2	Outlet pressure					0.45MPa+0.5MPa/6.1 GPM
8-3	Running current of each phase	L1:		L1:		11.0<A<17.0
		L2:		L2:		11.0<A<17.0
		L3:		L3:		11.0<A<17.0
8-4	Running voltage of each phase	L1&L2:		L1&L2:		(380/460)+/- 10%
		L1&L3:		L1&L3:		(380/460)+/- 10%
		L2&L3:		L2&L3:		(380/460)+/- 10%
9	Verifying proper operation of refrigerant system, record the following operating data .					
9-1	Compressor discharge pressure					<2400kpa
9-2	Condenser air inlet temperature					<45℃
9-3	Supply coolant temperature (read from PLC)					18.0℃<T<20.0℃, +/- 1℃
9-4	Return coolant temperature (Read from PLC)					24.5℃<T<28.5℃
9-5	Compressor suction pressure					>436kpa
9-6	Compressor Input voltage of each phase	L1&L2:		L1&L2:		(380/460)+/- 10%
		L1&L3:		L1&L3:		(380/460)+/- 10%
		L2&L3:		L2&L3:		(380/460)+/- 10%
9-7	Coil temperature (read from PLC)					<55 deg ℃/ 131 deg ℉
9-8	Compressor discharge pipe temperature at 4 to 8 inches from the discharge port.					<90 deg ℃/194 deg ℉
9-9	Compressor running current of each phase	L1:		L1:		5.0<A<10.0
		L2:		L2:		5.0<A<10.0
		L3:		L3:		5.0<A<10.0
9-10	Chiller total running current of each phase	L1:		L1:		11.0<A<17.0
		L2:		L2:		11.0<A<17.0
		L3:		L3:		11.0<A<17.0
9-11	Check refrigerant sight glass status (add refrigerant and do leak-test if it indicates a low refrigerant charge)					Refrigerant should be dry and no or only few bubble
10	Visual check and clean for the equipment.					record damages
11	Check PLC parameters setting by compareing with the table of parameters in user manual, and correct the wrong settings.					refer to parameter in manual
12	Verify that the cooling capacity meets the requirement of the GE equipment.Record the supply water temperature every 5 minutes for 30 minutes.			T1: T2: T3: T4: T5: T6:		19 ℃ +/- 1℃
13	Confirm the correct setting on the 50/60 Hz setting. See section 2.4 in user manual.					refer to manual, Section 2.4
14	Record the ambient temperature and temperature at inlet of condenser.					< 45 deg℃
15	Check the oil in the compressor. (can this be checked?)					Shall be at least half full.
16	Confirm the condenser fan is operating normally. Check the fan speed can be changed through obsevation.					
17	Check condenser coils for dirt or damage, clean the coil if necessary					
18	Confirm coolant supply temperature setting is 19 deg C.					
19	Record compressor inlet temperaure.					>0℃

Table- 13 Preventive check list

4.3 Consumption Parts and Spare Parts Order Information

S/N	Part Number	Description	Model	Unit	Designator
1	8.15.20.03600	Compressor	ZR54KCE-TFD-552	pcs	**COM**
2	8.68.10.02850	Pump	NMD 20/140AE	pcs	**PUMP**
3	8.25.10.02850	Axial fan	FE050-6EK.4F.6	pcs	**FAN**
4	8.45.25.01800	Transformer TC2,3	HLBB-0020-0003	pcs	**TC2,TC3**
5	8.45.50.02500	Masterpact	OT25E3	pcs	**QS**
6	8.45.45.04500	AC contactor	A12-30-10，230V	pcs	**KM1**
7	8.45.45.08010	AC contactor	A9-30-10，230V	pcs	**KM3**
8	8.45.55.16350	Circuit breaker	S272C6	pcs	**QF1**
9	8.45.55.16450	Circuit breaker	S271C4	pcs	**QF2**
10	8.45.60.34350	Auxiliary relay	HH52P-L 220V	pcs	**KA1**
11	8.45.85.22700	Terminal block	UK10N	pcs	
12	8.45.60.29920	Thermal relay	TA25DU11	pcs	**FR1**
13	8.45.60.29820	Thermal relay	TA25DU5	pcs	**FR3**
14	8.45.40.09770	Phase controller	CM-PFS	pcs	**FV1**
15	8.45.87.04350	Fuse	0.25A	pcs	**FU1**
16	8.45.85.13060	Fuse terminal block	UK5-HESI	pcs	
17	8.35.55.06760	Solenoid valve	EVR6	pcs	**YV**
18	8.35.55.07050	Coil for solenoid	230V,50/60Hz	pcs	**YV1**
19	8.15.54.11010	Thermal expansion valve	TDEZ4	pcs	**VCT**
20	8.35.50.01250	Filter-drier	EK164S	pcs	**FG**
21	8.68.30.01040	Water level switch	DHRE*BN02560	pcs	**WL1,WL2**
22	8.35.45.51200	Quick disconnection（male)	6603-12-12	pcs	
23	8.35.45.07600	Quick disconnection(female)	6603-12-12	pcs	
24	8.45.25.01700	Transformer TC1	650VA 50/60Hz	pcs	**TC1**
25	8.45.50.06100	Switch handle	OHB1AH1	pcs	**QS**
26	8.45.60.34340	Auxiliary relay	HH52P-L 12V	pcs	**KAP**
27	8.45.60.34800	Auxiliary relay socket		pcs	
28	8.45.45.11750	Assistant terminal	CAL5-11	pcs	**KM1**
29	8.45.50.05500	Prolonged axis	OXS5X120	pcs	**QS**
30	8.45.96.01360	Heater	018-0072-00	pcs	**F**
31	8.35.55.00600	Shut valve	3/8"	pcs	**VS**
32	8.45.11.00090	Remote controller		pcs	
33	8.45.11.00140	PC main controller	ENERGY ECH 211	pcs	
34	8.45.11.00080	CF-15 fan regulator	CF15	pcs	**CF-15**
35	8.45.11.00100	Sensor 1, ST3	cond. temp.sensor	pcs	**BP-T**
36	8.45.11.00110	Sensor 2, ST1&2	coolant temp. sensor	pcs	**BP-T**
37	8.35.10.01000	Sight glass	AMI-ISS4	pcs	**IPL**
38	8.45.40.05450	High/low pressure valve	DNS-D606X	pcs	**SP-L/SP-H**
39	8.45.40.05550	Temperature switch	ALS-C1090L1	pcs	**ST**
40	8.15.25.12050	Evaporator	CB26-36H	pcs	**E**
41	8.65.20.01600	Caster	DL314PPBGGS	pcs	
42	8.35.20.07100	Refrigerant receiver	76X235	pcs	**RL**
43	8.68.70.00850	Water pressure meter	1/2"	pcs	**MW**
44	8.68.70.01020	Y type water filter	DN20	pcs	**FA**
45	8.35.30.02350	Water relief valve	SVB35WS3	pcs	**YS**
46	8.68.70.05100	Metal Flexible tube	1" 475mm	pcs	
47	8.68.70.05300	Metal Flexible tube	1/2" 1000mm	pcs	
48	8.68.70.01680	Filler		pcs	**FA**
49	8.15.65.08850	Condenser	P-MEDI10	pcs	**C**
50	8.35.40.01260	Reservoir exceed valve	6110/22	pcs	
51	8.35.45.18010	Brass coupling for hose		pcs	
52	8.68.70.01030	Drain valve	1/2"	pcs	
53	8.35.45.43250	Parallel connector	1-1/4"	pcs	
54	8.35.45.29000	Connector	1/2"	pcs	
55	8.35.45.42250	Connector	3/4"	pcs	
56	8.68.70.01850	External Connector	1"x3/4"	pcs	
57	8.66.10.01150	Pump seal	R5H2-X7X7RZ7	pcs	
58	8.68.70.06060	Liquid level gauge	L480	pcs	
59	7.35.40.05860	Rubber hose	3/4" 100ft	set	

Table- 14 Spare Parts List

Section 5 – Troubleshooting

The controller can perform full system self-diagnose and displays alarms in serial code accordingly. In the event of a system failure, the alarm type will be shown on the display.

NOTE: When an alarm is displayed, the system will not cool the coolant.

When an alarm is triggered, two things occur:

1) The corresponding loads are shut down.

2) The alarm code appears on the display.

The alarm message consists of a code with the format "Exx" (where "xx" is a 2-digit number, indicating the type of failure detected, such as: E00, E25, E39….) all possible alarms are listed in the Table 14 – Alarm Codes List & Possible Solution, along with the codes the corresponding loads will be shut down.

Before you reset the current alarm, please check if any other alarms code exist following steps as below:

1) At the status of current alarms display, by pressing both mode and on off at the same time and then releasing within 2 seconds, you can move from level 0 deeper into level 1.

2) Pressing mode or on off, you can turn the page to the display of Err as shown in Fig- 25.

3) Press both mode and on off at the same time and then release within 2 seconds, and you move from level 1 deeper into level 2.

4) Pressing mode or on off, you can inquiry history alarms. The alarm is listed one by one in the order of descending priority.

All alarm codes would disappear when power is turned off.

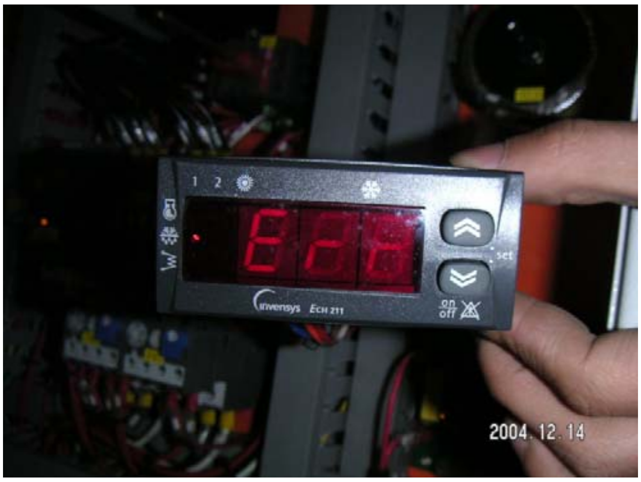


Fig.- 25 "Err" page

Pressing the ON-OFF button and releasing to manually reset alarm.

Manual reset shuts down corresponding loads and requires an operator to intervene (reset the alarm using the ON-OFF control).

NOTE: Manual reset are used mainly in case of problems which could result in damage the system. Therefore check the system to relieve the alarm before reset.

When the alarm occure, you should check the relevant parts per the "item/component check" description and be sure of the chiller is running under the right environmental conditions.

Check the status of the fuse, FV1, and the status of internal breakers, QF1 and QF2, if the controller doesnot come on when you switch on power.

Code	Loads Shut Down	Signal	Description	Possible Cause	Check Points	Recommend Actions
E00	Compressor Capacity Step Condenser Fan Pump	Remote Off or the outlet water temperature exceeds 53°C	All loads will be shut down; Only show on the control panel LCD when remote controller off reset automatically when the alarm issue recovered	1.coolant outlet temperature exceeds 53C degree(127.4F)	check whether E02 alarm code is recorded; if yes, then the unit didn't cool the coolant	first solve the problem according to the alarm E02
				2. low coolant flow rate causes pressure of compressor drop, which leads to refrigerant suction; as result, the refrigeration capacity becomes smaller and the coolant temperature becomes too high	check if there is blockage in the coolant loop and adjust flow rate if necessary	adjust or clean the coolant loop or replace the coolant filter if necessary
				3.loose wire in the high temperature protector	check if this is the case	tighten up the line or reconnect line in the component
E04 (1)	Compressor Condenser Fan	Compressor high pressure alarm, or compressor thermal switch protection, or fan thermal protection.	Compressor and fan will be shut down. water pump continues to run	1. compressor running current exceed the setting point of the current protector	Check power supply and running currents of the compressor; Check current setting.	Correct the current setting and power supply.
				2. the high pressure of the refrigeration loop exceeds setting point(29bar/420.5psi)	a. Is the fan speed regulator wire loose? If so, the condenser may not be cooled properly.	fasten the loose wire or connections of these terminals
					b. Is the condenser with dirt or something	clean the condenser and restore the desired cooling effect
					c. the environment temperature is higher than 43C degree(109.4F)	the operating condition exceeds the design spec, pls contact the Field engineer or tech support
					d. Debris in sealed system clogging TEV or refrigerant filter	clean the refrigeration sub-system and recharge the refrigerant or replace the filter if necessary
				3. the fan thermal protector activation	a. Is the condenser covered by flyers?	remove the covers on the condenser
					b. Is the exhaust temperature higher than setting point?	the operating condition exceeds the design spec, pls contact the Field engineer or tech support
					c. Is anything blocking the fan?	clean the fan
				4.In the controller, the fan speed/parameter setting is incorrect	check the parameter in the controller	correct the wrong parameters
				5.the capacity of the fan failed	Measure the capacity of the component to verify if it is ok	Replace it with a new one
				6. loose wire in high pressure protector	high/low pressure protector	tighten up the loose wire or reconnect the line of these terminal

Code	Loads Shut Down	Signal	Description	Possible Cause	Check Points	Recommend Actions
				7. high/low pressure protector failed	simulate the pressure to activate the protector.	if the component failed the test, replace the component.
				8. fan motor(or thermal overload protector in the fan motor) or fan motor start capacitance failed	measure the power input the fan, if input power is ok , the capacitance or the fan motor maybe fail.	connect the service engineer to replace the axial fan or the start capacitance.
				9. everything is ok but the fan leaves was stalled by something	check the fan leaves if it could rotate freely by hand	remove the things that stalled the fan leaves
E04 (2)	Compressor Capacity Step Condenser Fan Pump	Thermal switch protection pump	All loads will be shut down; Active if the current of pump exceed 4.5A.;Automatically reset unless alarm events per hour reaches the value of parameter Pa A09 (2), after which manually reset;	1.water pump running current exceeds the setting point of the current protector	a.Is pump electrical resistance normal?	If so, replace the whole pump or the motor
					b. Is there something stuck in the pump impeller? check the current or the noisy of the motor when it is running	If so, remove the blockage
					c. Is the pump rusty?	If so, replace the pump and clean the water loop
				2.loose wire in the pump thermal protector(FR4)	Is the line connection in the pump thermal protector good?	tighten up the line connection
E02	Compressor Capacity Step Condenser Fan Pump	Low pressure (digital) or high water level	Compressors and condenser fan will be shut down Active if low pressure (suction pressure) below 0.6Bar when the chiller is operating or the water leak, water level in tank below the high level when the chiller is in stand by.	1.coolant leak in pipe system; high coolant level in water reservoir activated, the chiller wouldn't cool the coolant	a. every pipe joint in the chiller ,pump seal	properly seal the leak spots
					b. Is high coolant level switch failed	replaced the switch
				2. refrigerant leak in refrigeration loop, compressor suction pressure is lower than the setting point 0.5bar(7.3psi)	check all the refrigeration loop	properly seal the leak point
				3.TEV failed and cause the suction pressure too low	check suction pressure when the chiller still running	replace TEV
				4. blockage in refrigeration system and suction pressure too low	check suction pressure when unit still running, check if the coolant temperature is correct	replaced TEV and clean the refrigerant system
				5. false alarm due to the failure of high/low pressure protector switch	simulate the pressure to activate the protector.	replace the protector if the protector fails to respond
				6. loose wiring triggers the alarm	check all the line connection about this alarm according to the electrical diagram	tighten up the connection or reconnect the terminals
				7. low ambient temperature	does the environment temperature drop below -30C degree(-22F)	the operating condition is outside of specified range, pls contact the Field engineer or tech support

Code	Loads Shut Down	Signal	Description	Possible Cause	Check Points	Recommend Actions
				2.loose wire in the pump thermal protector(FR4)	Is the line connection in the pump thermal protector good?	tighten up the line connection
				3. the thermal overload relay of the pump failed	the thermal overload relay	replace the component with new one
E05	Compressor Capacity Step Condenser Fan	Anti-freeze	Condenser fan and compressors will be shut down; Active if the return water temperature below Pa A11 (-31°C); Goes off if it is greater than Pa A11 + Pa A12 (-30°C); Automatically reset unless alarm events per hour reaches the value of parameter Pa A13 (20), after which manually reset;	1. ambient temperature of the chiller exceed the running range of the unit	measure the ambient the temperature of the chiller	if the temperature lower than -31C degree(-23.8F),pls contact the field engineer or tech support
				2. parameters in controller is setting incorrectly	check the parameter A11,A12,A13 right or not	refer to the user manual and correct the parameters according to the parameter table
				3. the temperature of the ST3(condenser temperature probe) is incorrect	compare with the standard temperature sensor to judge if the sensor is good	replace the failed sensor
E06	Compressor Capacity Step Condenser Fan Pump	Probe ST2 fault	All loads will be shut down; Triggered if the return water temperature sensor shorts or is cut off or probe limits are exceeded (-50°C.. 100°C).	1. the temperature sensor ST2 failed	check the resistance of the sensor	replace the sensor
				2. loose wire at the terminal block	terminal block in electrical box	tighten up the line or reconnect the terminal block
E07	Compressor Capacity Step Condenser Fan Pump	Probe ST3 fault	All loads will be shut down; Triggered if condensing temperature sensor shorts or is cut off or probe limits are exceeded (-50°C.. 100°C).	1. the temperature sensor ST3 failed	check the resistance of the sensor	replace with a new one
				2. loose wire in the terminal block	terminal block in electrical box	tighten up the line or reconnect the terminal block
E11	Compressor Capacity Step	High condensation temperature (analogue)	Compressors will be shut down; Active if condensing temperature sensor detects a value in excess of Pa A14 (65°C) Turned off if temperature falls below Pa A14 – Pa A15 (50°C). Always manually reset	1.ambient temperature of the chiller exceeds the operation range of the unit	measure the ambient temperature of the chiller	if the temperature EXCEED 65C degree(149F),pls contact the manufacture
				2.parameters in controller is set incorrectly.	check the parameter A14,A15 right or not	refer to the user manual and set the parameters according to the parameter table
				3.the temperature of the ST3(condenser temperature probe) is incorrect	compare with the standard temperature sensor to judge if the sensor is good	replace the bad one
				4. the fan speed regulator is wrong	replace with a new one the check if the component is failed	if the component was wrong ,replace it by service engineer
				5.fan motor (or thermal overload protector in the fan motor)or fan motor start capacitance failed	measure the power input the fan, if input power is ok , the capacitance or the fan motor maybe fail.	contact the service engineer to replace the axial fan or the start capacitance.
				6. parameters in controller about fan is set incorrectly.	check all the parameters in controller of "fan"	correct the wrong parameters
				7. everything is ok but the fan leaves was stalled by something	check the fan leaves if it could rotate freely by hand	remove the things that stalled the fan leaves

Code	Loads Shut Down	Signal	Description	Possible Cause	Check Points	Recommend Actions
E12	Compressor Capacity Step Condenser Fan	low condensation temperature (analogue)	Compressors and fans will be shut down; Active if condensing temperature sensor detects a value below Pa A17 (-50°C) Goes off if temperature/pressure exceeds Pa A17 + Pa A18 (-45°C). Automatically reset unless alarm events per hour reaches the value of parameter Pa A19 (20), after which manually reset; Inactive during timer Pa A16 (10s) after compressor is turned on	1. ambient temperature of the chiller exceeds the operation range of the unit	measure the ambient the temperature of the chiller	if the temperature lower than -50C degree(-58F),pls contact the manufacture
				2. parameters in controller is setting wrong	check the parameter A17,A18,A19 right or not	refer to the user manual and correct the parameters according to the parameter table
				3. the temperature of the ST3(condenser temperature probe) is wrong	compare with the standard temperature sensor to judge if the sensor is good	replace the bad one
E40	Compressor Capacity Step Condenser Fan Pump	Probe ST1 fault	All loads will be shut down; Triggered if the outlet water temperature sensor shorts or is cut off or probe limits are exceeded (-50°C.. 100°C).	the temperature sensor ST1 failed loose wire at the terminal block	1.check the resistance of the sensor 2.check terminal block in electrical box	replace the sensor tighten up the line or reconnect the terminal block
E41	Compressor Capacity Step Condenser Fan Pump	Low water level	All compressors, external fans and pumps will be turned off if the alarm is to be manually reset; Triggered if the low water level switch active for an amount of time equal to Pa A04 (10s); Goes off if the low water level switch remains inactive for an amount of time equal to Pa A05 (5s); Automatically reset unless alarm events per hour reaches the value of parameter Pa A06 (1), after which manually reset; Inactive during timer Pa A03 (5s) after pump (hydraulic pump) is turned on	2. loose wire at the terminal block	terminal block in electrical box	tighten up the line or reconnect the terminal block
					b. is low coolant level switch failed ?	replaced the switch
				2. parameters in controller is set incorrectly	check the parameter A03,A04,A05,A06 right or not	refer to the user manual and correct the parameters according to the parameter table
				3. loose wire between the level switch and terminal block	check line connecting of water level switch and terminal block	Tighten up the line connection
E42	Compressor Capacity Step Condenser Fan	ST1 temperature sensor failed	Active if the temperature exceed 100C(212F) or lower than –50C(-58F) and if the sensor was shorted or cut off	1. loose wire on the terminal block	check line connecting of the temperature sensor on terminal block	tighten the loosed wire
				2.the temperature sensor of the ST1(water outlet temperature probe) failed	compare to the standard temperature sensor to judge if the sensor is good	replace the failed sensor

Code	Loads Shut Down	Signal	Description	Possible Cause	Check Points	Recommend Actions
E46	Compressor Capacity Step Condenser Fan Pump	Over-temperature	Compressors will be shut down Triggered if probe ST1 (refer to analogue inputs) has a value over Pa A25(50°C) for an amount of time in excess of Pa A26(5s); Signals high temperature/pressure without shutting down the machine. These conditions are not sufficient to trigger the pressure switch, but can cause premature wear or damage to the machine.	1. all the other possible conditions causing compressor shut off and water temperature go up	check the controller to find out other alarm codes registered earlier	to deal with issues corresponding to the relevant Exx Code first as described in earlier error code sheet
				2. ambient temperature of the chiller exceed the design range of the unit	measure the ambient the temperature of the chiller	if the temperature EXCEED 65C degree(149F),pls contact the field engineer or tech support
				3. parameters in controller is set incorrect	check the parameter A25,A26 right or not	refer to the user manual and set the parameters according to the parameter table
				4. the temperature sensor of the ST1(condenser temperature probe) failed	compare with the standard temperature sensor to judge if the sensor is good	replace the bad one

Table- 15 Alarm Codes List & Possible Solution

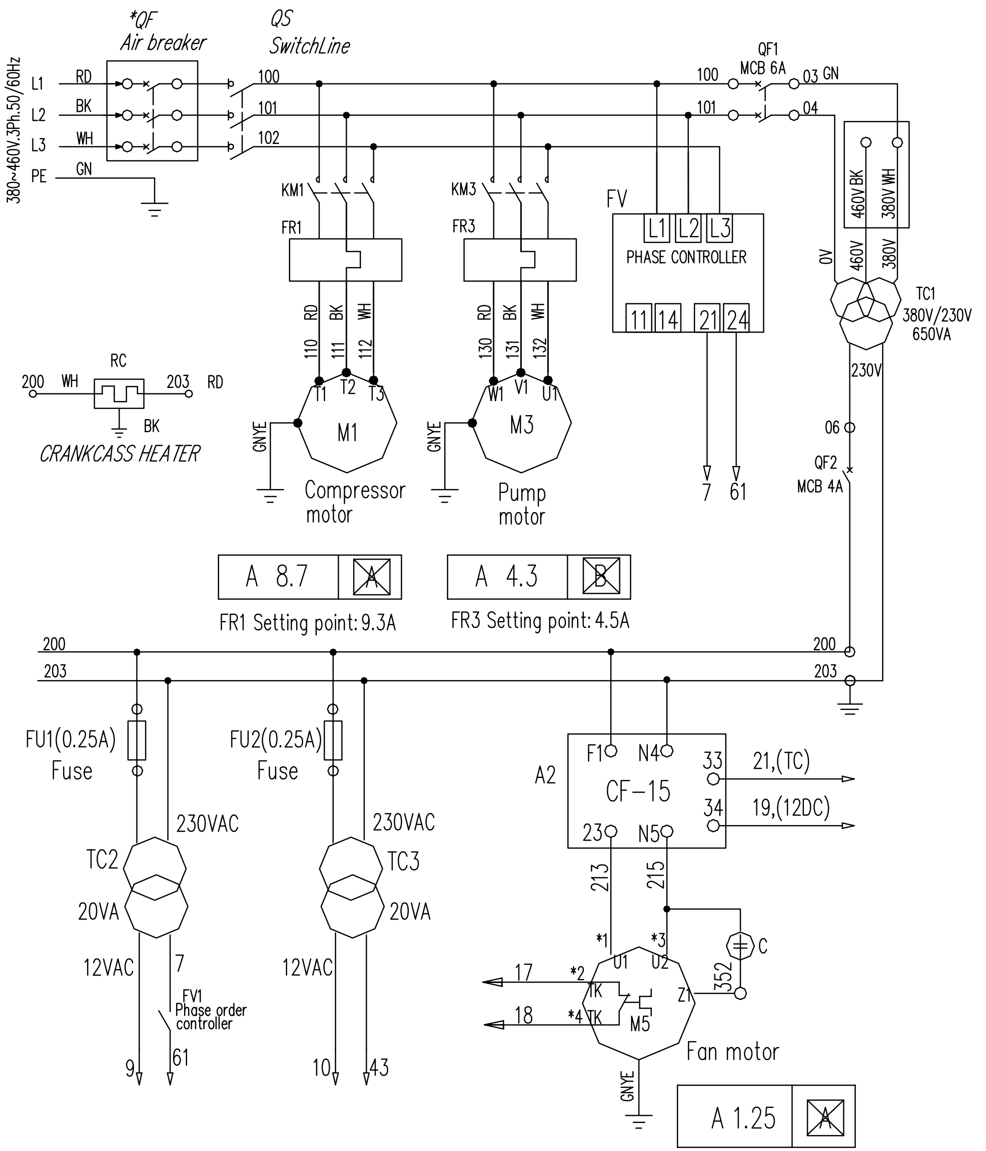
APPENDIX A – Table of Parameters

CONFIGURATION PARAMETERS				
Par.	Description	Value	Limits	Unit of measurement
H01	Maximum *set point* during *heating*	60	H02 - 90.0	°C
H02	Minimum *set point* during *heating*	30	-40.0 - H01	°C
H03	Maximum *set point* during *cooling*	25	H04 - 90.0	°C
H04	Minimum *set point* during *cooling*	15	-40.0 - H03	°C
H05	Configuration ST1	1	0 - 3	Num
H06	Configuration ST2	1	0 - 3	Num
H07	Configuration ST3	1	0 - 5	Num
H08	Configuration ST4	2	0 - 3	Num
H09	Bottom of scale pressure value	300	0 - 300	Kpax10
H10	Polarity ID1	1	0 - 1	Flag
H11	Polarity ID2	1	0 - 1	Flag
H12	Polarity ID3	1	0 - 1	Flag
H13	Polarity ID4	0	0 - 1	Flag
H14	Polarity ID5	1	0 - 1	Flag
H15	Polarity ST1	1	0 - 1	Flag
H16	Polarity ST2	1	0 - 1	Flag
H17	Polarity ST4	0	0 - 1	Flag
H18	Configuration ID3	1	0 - 4	Num
H19	Configuration ID4	1	0 - 4	Num
H20	Configuration ID5	2	0 - 4	Num
H21	Configuration ST4 if digital input	4	0 - 4	Num
H22	Configuration relay 2	0	0 - 1	Num
H23	Configuration relay 3	2	0 - 1	Num
H24	Configuration relay 4	0	0 - 2	Num
H25	Configuration of *optional output*	0	0 - 3	Num
H26	Configuration of serial protocol (not used)	1	0 - 1	Num
H27	Selection of operating mode	0	0 - 2	Num
H28	Presence of heat pumpt	0	0 - 1	Flag
H29	Heating mode set point	10	0 - 255	°C
H30	Mode selection differential	15	0 - 25.5	°C
H31	Enable dynamic set point	1	0 - 1	Flag
H32	Dynamic set point offset in cooling mode	1.5	0 - 25.5	°C
H33	Dynamic set point offset in heating mode	5	0 - 25.5	°C
H34	Outdoor temperature *set point* in *cooling* mode	13	0 - 255	°C
H35	Outdoor temperature *set point* in *heating* mode	22	0 - 255	°C
H36	Outdoor temp. ***dynamic set point*** differential in **cooling**	5	0 - 25.5	°C
H37	Outdoor temp. **dynamic set point** differential in **heating**	5	0 - 25.5	°C
H38	**Reversing valve** polarity	0	0 - 1	Flag
H39	Offset ST1	0	-12.7 - 12.7	°C
H40	Offset ST2	0	-12.7 - 12.7	°C
H41	Offset ST3	0	-127 - 127	°C/10 – Kpax10
H42	Offset ST4	0	-12.7 - 12.7	°C/10
H43	Mains frequency	0	0 – 1	Flag
H44	Family serial address	0	0 - 14	Num
Par.	Description	Value	Limits	Unit of measurement
H45	Device serial address	0	0 - 14	Num
H46	User password	1	0 - 255	Num
H47	**Copy card** write password	1	0 - 255	Num
H48	Number of **compressors** per circuit	2	1 – 2	Num
H49	Enable pressure/temperature based operation	3	0 – 2	Num
H50	Compressor on sequence	1	0 – 1	Num
H51	Compressor 2 or capacity step polarity	1	0 – 1	Num
H52	Selection of degrees °C or °F	0	0 – 1	Num

ALARM PARAMETERS				
Par.	Description	V a l u e	Limits	Unit of measurement
A01	Low-pressure switch bypass time after comp. on	60	0 - 255	Seconds
A02	Low pressure alarm events per hour	4	0 - 255	Num
A03	Bypass low level switch after pump on	5	0 - 255	Seconds
A04	Duration of active low level switch input	10	0 - 255	Seconds
A05	Duration of inactive low level switch input	5	0 - 255	Seconds
A06	Number of low level switch *alarm events per hour*	1	0 - 255	Num
A07	Compressor thermal switch bypass following comp. on	5	0 - 255	Seconds
A08	Compressor 1/2 thermal switch *alarm events per hour*	2	0 - 255	Num
A09	Fan thermal switch alarm events per hour	3	0 - 255	Num
A10	Anti-freeze alarm bypass after ON-OFF	5	0 - 255	Minutes
A11	Anti-freeze alarm *set point*	-31	-127 - 127	°C
A12	Anti-freeze alarm *hysteresis*	1	0 - 25.5	°C/10
A13	Anti-freeze alarm events per hour	20	0 - 255	Num
A14	Analogue input high pressure *set point*	650	0 - 900	°C/10 – Kpax10
A15	Analogue input high pressure *hysteresis*	50	0 - 255	°C/10 – Kpax10
A16	Analogue input low pressure bypass	10	0 - 255	Seconds
A17	Analogue input low pressure *set point*	-500	-500 - 800	°C/10–Kpax10
A18	Analogue input low pressure *hysteresis*	50	0 - 255	°C/10 – Kpax10
A19	Analogue input low pressure **alarm events per hour**	20	0 - 255	Num
A20	Machine out of coolant differential	30	0 - 25.5	°C/10
A21	Machine out of coolant bypass	120	0 - 255	Minutes
A22	Machine out of coolant duration	120	0 - 255	Minutes
A23	Machine out of coolant alarm activation	0	0 - 1	Flag
A24	Enable low pressure alarm during **defrosting**	0	0 - 1	Flag
A25	Over-temperature **set point**	50	0 - 255	°C
A26	Over-temperature ON duration	10	0 – 155	Seconds
COMPRESSOR PARAMETERS				
Par.	Description	V a l u e	Limits	Unit of measurement
C01	ON-OFF safety time	6	0 - 255	Secondsx10
C02	ON-ON safety time	18	0 - 255	Secondsx10
C03	*Cooling* regulation algorithm *hysteresis*	0.8	0 - 25.5	°C
C04	*Heating* regulation algorithm *hysteresis*	2	0 - 25.5	°C
C05	Regulation algorithm step intervention differential	1.0	0 - 255	°C
C06	Compressor 1 – compressor 2 (step) on interval	60	0 - 25.5	Seconds
C07	Compressor 1 – compressor 2 (step) off interval	3	0 - 25.5	Seconds
FAN CONTROL PARAMETERS				
Par.	Description	V a l u e	Limits	Unit of measurement
F01	Fan output configuration	0	0 - 2	Num
Par.	Description	V a l u e	Limits	Unit of measurement
F02	Fan *pick-up* time	0	0 - 255	Seconds/10
F03	Fan phase shift	4	0 - 100	%
F04	Impulse duration of triac on	10	0 - 255	uSx10
F05	Functioning in response to compressor request	0	0 - 1	Flag
F06	Minimum speed during *cooling*	30	0 - 100	%
F07	Silent speed during cooling	90	0 - 100	%
F08	Minimum fan speed temperature/pressure *set point* during *cooling*	200	-500 - 800	°C/10 – Kpax10
F09	Prop. band during *cooling*	200	0 - 255	°C/10 – Kpax10
F10	*Cut-off* differential	100	0 - 255	°C/10 – Kpax10
F11	Cut-off hysteresis	80	0 - 255	°C/10 – Kpax10
F12	*Cut-off* bypass time	20	0 - 255	Seconds
F13	Maximum speed during *cooling*	100	0 - 100	%
F14	Maximum fan speed temperature/pressure *set point* in *cooling* mode	450	-500 - 800	°C/10 – Kpax10
F15	Minimum speed during *heating*	60	0 - 100	%
F16	Silent speed during heating	100	0 - 100	%

F17	Minimum fan speed temperature/pressure *set point* during *heating*	350	-500 - 800	°C/10 – Kpax10
F18	Proportional band during *heating*	100	0 - 255	°C/10 – Kpax10
F19	Maximum speed during *heating*	100	0 - 100	%
F20	Maximum fan speed temperature/pressure **set point** during **heating**	300	-500 - 800	°C/10 – Kpax10
F21	**Internal fan** step differential	2	0 - 25.5	°C
F22	**Internal fan** step **hysteresis**	1	0 - 25.5	°C
F23	Hot start **set point**	50	0 - 255	°C
F24	Hot start **hysteresis**	1	0 - 25.5	°C
F25	Preventilation in **cooling** mode	5	0 - 25.5	Seconds
PUMP PARAMETERS				
Par.	Description	V a l u e	Limits	Unit of measurement
P01	Pump operating mode	0	0 - 1	Flag
P02	Delay between pump ON and compressor ON	20	0 - 255	Seconds
P03	Delay between compressor OFF and pump OFF	5	0 - 255	Seconds
ANTI-FREEZE/BOILER PARAMETERS				
Par.	Description	V a l u e	Limits	Unit of measurement
r01	Configuration of electrical heaters in defrost mode	1	0 - 1	Flag
r02	Configuration of electrical heaters on in *cooling* mode	0	0 - 1	Flag
r03	Configuration of electrical heaters on in *heating* mode	0	0 - 1	Flag
r04	Configuration of anti-freeze electrical heater control probe in *heating* mode	1	0 - 1	Flag
r05	Configuration of anti-freeze electrical heater control probe in *cooling* mode	1	0 - 1	Flag
r06	Configuration of electrical heaters when OFF or on *stand-by*	1	0 - 1	Flag
r07	*Set point* of anti-freeze electrical heaters in *heating* mode	20	Par. r09 – Par. r10	°C
r08	*Set point* of anti-freeze electrical heaters in *cooling* mode	10	Par.r09 –Par.r10	°C
r09	Maximum *set point* of anti-freeze electrical heaters	90	Par. r10 – 127	°C
Par.	Description	V a l u e	Limits	Unit of measurement
r10	Minimum *set point* of anti-freeze electrical heaters	10	-127– Par.r09	°C
r11	Anti-freeze heater *hysteresis*	1	0 - 25.5	°C
r12	Set point of external anti-freez**e electrical heaters**	10	Par. r09 – Par. r10	°C
r13	Outdoor temperature **set point** for **boiler** on	10	-127 - 127	°C
r14	Outdoor temperature differential for **boiler** off	2	0 - 25.5	°C
DEFROST PARAMETERS				
Par.	Description	V a l u e	Limits	Unit of measurement
d01	Defrost enabled	1	0 - 1	Flag
d02	*Defrost start* temperature/pressure	-20	-500 - 800	°C/10 – Kpax10
d03	Defrost interval (response time)	20	0 - 255	Minutes
d04	*Defrost end* temperature/pressure	300	-500 - 800	°C/10 – Kpax10
d05	Maximum defrost time	30	0 - 255	Minutes
d06	Compressor-*reversing valve* wait time	30	0 - 255	Seconds
d07	Drip time	30	0 - 255	Seconds
d08	Temperature at which ***defrost start***s if **Pa H49**= 1	8	-50.0 - 80.0	°C/10
d09	Temperature at which **defrost end**s if **Pa H49**= 1	28	-50.0 - 80.0	°C/10

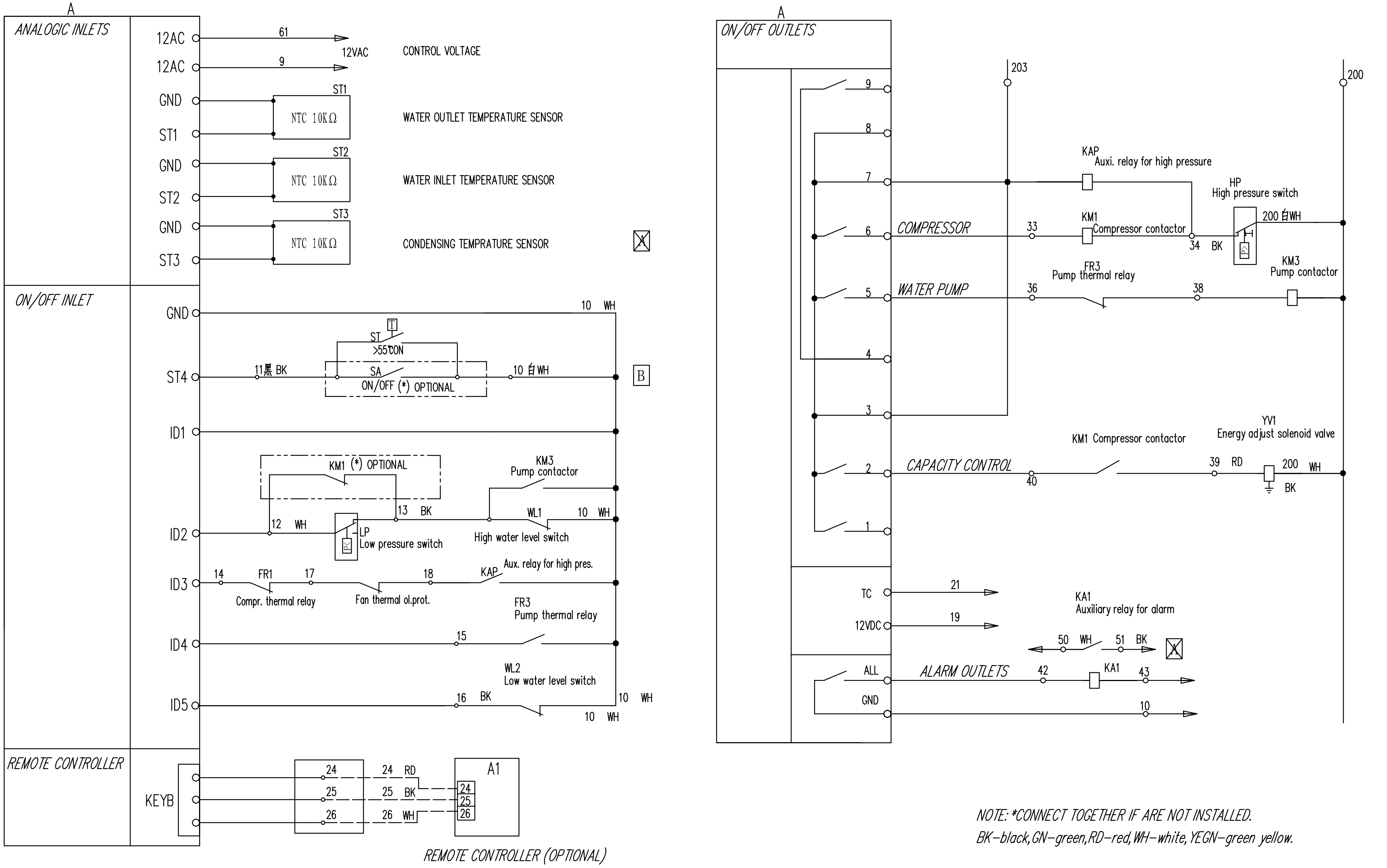
APPENDIX B – Wiring Diagrams Circuit Diagram (1)



21	FUI、FU2	Fuse 0.25A	4	
20		Terminals UK5-HESI	2	
19	KAP	Relay HH52/220VAC	1	
18	KA1	Relay HH52/12VAC	1	
17	XT	Terminal board GE1	1	
16	ST3	Temp. sensor SN691150	1	INVENSYS
15	ST1、ST2	Temp. sensor SN8S0A1500	2	INVENSYS
14	A2	Fan Controller MW991100	1	INVENSYS
13	A1	Remote Controller MW320600	1	INVENSYS
12	A	Controller ECH211	1	INVENSYS
11	TC2、TC3	Transformer HLBB-0020-0003	1	
10	TC1	Transformer 650VA 50/60Hz	1	DONGAN USA
9		Auxiliary contact blocks CAL5-11	1	ABB
8	FV	Phase controller CM-PFS	1	ABB
7	FR3	Thermal overload relayTA25DU5	1	ABB
6	FR1	Thermal overload relayTA25DU11	1	ABB
5	QF2	MCB S271-C4	1	ABB
4	QF1	MCB S272-C6	1	ABB
3	QS	SwitchLine OT25E3	1	ABB
2	KM3	Contactor A9-30-10	1	ABB
1	KM1	Contactor A16-30-10	1	ABB
NO.	Designator	Description	QTY.	Remark

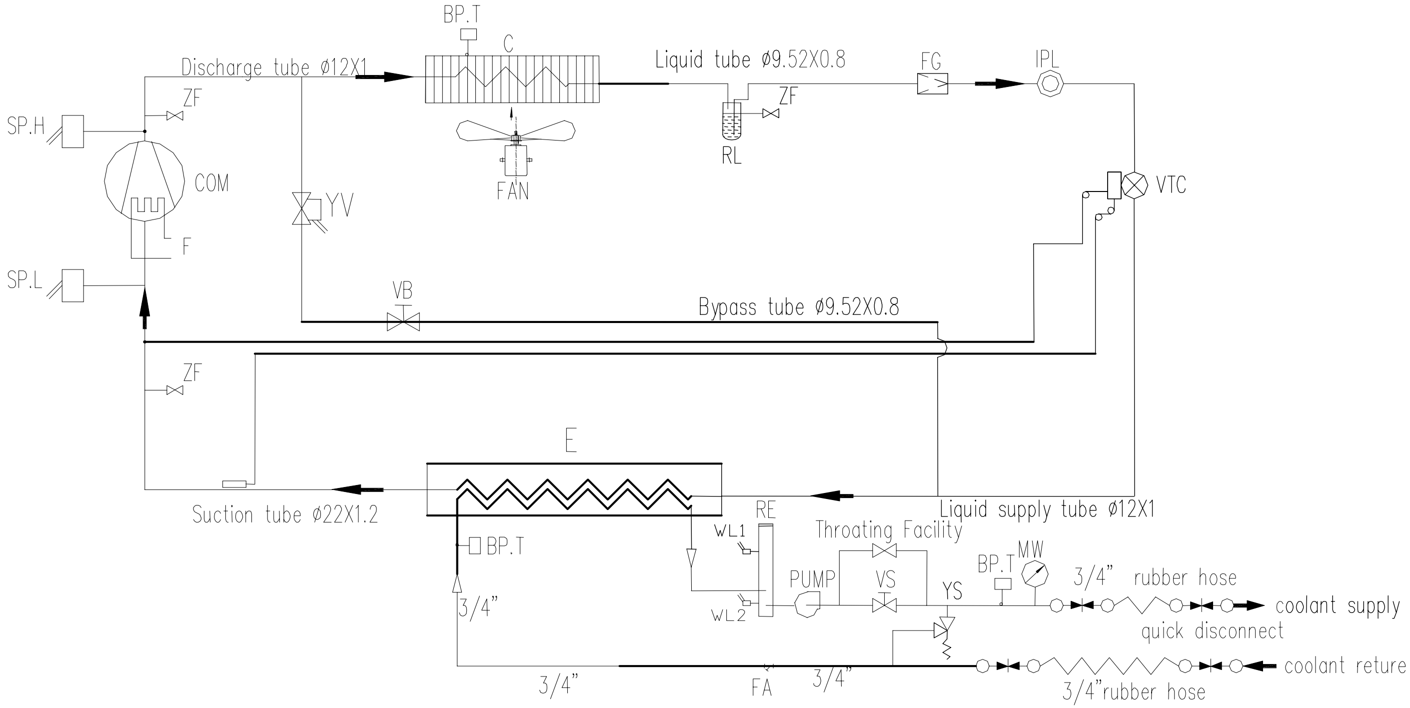
NOTES: * To install by customer

Circuit Diagram (2)



NOTE: *CONNECT TOGETHER IF ARE NOT INSTALLED.
BK-black, GN-green, RD-red, WH-white, YEGN-green yellow.

Appendix C: Piping Diagram



SP.L	低压压力开关/LOW PRESSURE SWITCH	ZF	针阀/NEEDLE VALVE
FG	制冷剂干燥过滤器/FILTER AND DRIER	RE	水箱/RESERVIOR
E	板式蒸发器/EVAPORATOR	VB	旁通调节阀/BYBASS ADJUSTMENT VALVE
VTC	膨胀阀/EXPANSION VALVE	YV	电磁阀/SOLENOID VALVE
IPL	示液镜/LIQUID INDICATOR	YS	安全阀/RELIEF VALVE
RL	储液器/LIQUID RECEIVER	FA	水过滤器/WATER FILTER
FAN	冷凝风机/CONDENSER FAN	MW	水压表/WATER PRESSURE GAUGE
F	曲轴箱加热器/CRANKCASE HEATER	BP.T	温度传感器/TEMPERATURE SENSOR
C	风冷冷凝器/CONDENSER	PUMP	水泵/WATER PUMP
SP.H	高压压力开关/HIGH PRESSURE SWITCH	VS	调压闸阀/SERVICE VALVE
COM	压缩机COMPRESSOR	WL	水位开关/WATER LEVEL SWITCH

Appendix D: Safety Data Sheet

Material Safety Data Sheet of Propylene Glycol

Safety Data Sheet of R407C MSDS

MATERIAL SAFETY DATA SHEET

PROPYLENE GLYCOL INDUSTRIAL

MSDS No P000029-1-ACAP-AE
Ver. No 2
Ver. Date JAN 5 99

LYONDELL

Lyondell Asia Pacific, Ltd.
41st Floor, The Lee Gardens,
33 Hysan Avenue
Causeway Bay
Hong Kong

IMPORTANT: Read this MSDS before handling and disposing of this product and pass this information on to the employees, customers, and users of this product. This product is covered by the OSHA Hazard Communication Rule in the USA and this document has been prepared in accord with the MSDS requirements of this rule.

1. General

Trade Name	PROPYLENE GLYCOL INDUSTRIAL	**Telephone Numbers:** EMERGENCY (886) 933-635-556 CUSTOMER SERVICE (852) 2882-2668	
Other Company Names	PG		
Synonyms	None		
Other Industry Names	1,2-Propanediol; 1,2-Dihydroxypropane		
Chemical Family	Glycols	**IMO Proper Shipping Name** Not regulated **IATA Proper Shipping Name** Not regulated	
Generic Name	Monopropylene Glycol	**Transport Hazard Classification** **ADR/VLG** Not regulated **IATA** Not regulated **RID/VSG** Not regulated	**ADNR/VBG** Not regulated **IMO** Not regulated
CAS No. (See Section 9 - Components)	**Company Material ID** BE130	**MSDS Class** F	**UN/NA ID No.** Not Classified

2. Summary of Hazards

Signal Word	CAUTION
Physical Hazards	Aqueous solutions may produce flammable vapors Slightly combustible liquid
Acute Health Effects (Short-Term)	No inhalation hazard identified from data available; Slight eye irritant; No ingestion hazard identified from data available; No skin irritation hazard identified from data available; No skin absorption hazard identified from data available
Chronic Health Effects (Long-Term)	No chronic health hazards are expected to occur from anticipated conditions of normal use of this material

3. Fire and Explosion

Flash Point ~ 109°C (PMCC)	**Autoignition Temperature** ~ 371°C	**Flammable Limits** (at Normal Atmospheric Temp and Pressure) Lower: ~ 2.4 (% vol in air) Upper: ~ 17.4 (% vol in air)

Fire and Explosion Hazards	Heat from fire can generate flammable vapor. When mixed with air and exposed to ignition source, vapors can burn in open or explode if confined. Vapors may travel long distances along the ground before igniting and flashing back to vapor source. Fine sprays/mists may be combustible at temperatures below normal flash point. Aqueous solutions containing less than 95% propylene glycol by weight have no flash point as obtained by standard test methods. However aqueous solutions of propylene glycol greater than 22% by weight, if heated sufficiently, will produce flammable vapors. Only aqueous solutions of propylene glycol less than 22% should be used in sprinkler systems or other fire-fighting equipment. Always drain and flush systems containing propylene glycol with water before welding or other maintenance.
Extinguishing Media	Alcohol type foam CO_2 Dry chemical
Extinguishing Media Use Comment	Use waterspray/waterfog for cooling
Special Firefighting Procedures	Do not enter fire area without proper protection. Fight fire from a safe distance/protected location. Heat may build enough pressure to rupture closed containers/spreading fire/increasing risk of burns/injuries. Use water spray/fog for cooling. Avoid frothing/steam explosion. Burning liquid may float on water. Although water soluble, may not be practical to extinguish fire by water dilution. Notify authorities immediately if liquid enters sewer/public waters.

4. Health Hazards

Summary of Acute Hazards	Not expected to present a significant acute health hazard upon short term exposure.	
ROUTE OF EXPOSURE	**SIGNS AND SYMPTOMS**	**PRIMARY ROUTE(S)**
Inhalation	No significant signs or symptoms indicative of any adverse health hazard are expected to occur as a result of inhalation exposure.	No
Eye Contact	May cause minor eye irritation.	Yes
Skin Absorption	No significant signs or symptoms indicative of any health hazard are expected to occur as a result of skin absorption exposure.	No
Skin Irritation	No significant signs or symptoms indicative of any adverse health hazard are expected to occur as a result of skin exposure.	No
Ingestion	No significant signs or symptoms indicative of any health hazard are expected to occur as a result of ingestion.	No
Summary of Chronic Hazards	No adverse chronic health effects are expected from anticipated conditions of normal use of this material, unless aerosol is generated.	
Special Health Effects	This material or its emissions may aggravate pre-existing eye disease.	

5. Protective Equipment and Other Control Measures

Respiratory	No special respiratory protection is recommended under anticipated conditions of normal use with adequate ventilation.
Eye	Eye protection such as chemical splash goggles and/or face shield must be worn when possibility exists for eye contact due to splashing or spraying liquid, airborne particles, or vapor. Contact lenses must not be worn.
Skin	Not normally considered a skin hazard. Where use can result in skin contact, practice good personal hygiene. Wash hands and other exposed areas with mild soap and water before eating, drinking, smoking, and when leaving work.
Engineering Controls	No special ventilation is recommended under anticipated conditions of normal use beyond that needed for normal comfort control.
Other Hygienic Practices	Use good personal hygiene practices. Wash hands before eating, drinking, smoking, or using toilet facilities. Promptly remove soiled clothing/wash thoroughly before reuse. Shower after work using plenty of soap and water.
Other Work Practices	No special work practices are needed beyond the above recommendations under anticipated conditions of normal use.

6. Occupational Exposure Limits

Substance	Source	Date	Type	Value/Units	Time	Skin

No occupational exposure limit(s) have been established for this material or its components

Exposure Limit Comments	No additional Occupational Exposure Limit information available

7. Emergency and First Aid

Inhalation	Not expected to present a significant inhalation hazard under anticipated conditions of normal use.
Eye Contact	In case of eye contact, immediately rinse with clean water for 20-30 minutes. Retract eyelids often. Obtain emergency medical attention if pain, blinking, tears or redness persist.
Skin Contact	Not expected to present a significant skin hazard under anticipated conditions of normal use.
Ingestion	Not expected to present a significant ingestion hazard under anticipated conditions of normal use.
Physician's Emergency Medical Treatment Procedures	Treat symptomatically. Treatment of overexposure should be directed at the control of symptoms and the clinical condition of the patient. After adequate first aid, no further treatment is required unless symptoms reappear.
Physician's Detoxification Procedures	No detoxification information available.

8. Spill and Disposal

Precautions if Material is Spilled or Released
May contaminate water supplies/pollute public waters. Evacuate/limit access. Equip responders with proper protection. Prevent flow to sewer/public waters. Stop release. Notify fire and environmental authorities. Restrict water use for cleanup. Slippery walking. Spread granular cover. Impound/recover large land spill. Soak up small spills with inert solids. Use suitable disposal containers. On water, material is soluble and may float or sink. May biodegrade. Contain/collect rapidly to minimize dispersion. Disperse residue to reduce aquatic harm. Report per regulatory requirements.

Waste Disposal Methods
Contaminated product, soil, or water should not be designated hazardous waste. Landfill solids at permitted sites. Use registered transporters. Burn concentrated liquids, diluting with clean, low viscosity fuel. Avoid flame-outs. Assure emissions comply with applicable regulations. Dilute aqueous waste may biodegrade. Avoid overloading/poisoning plant biomass. Assure effluent complies with applicable regulations.

9. Components

(This may not be a complete list of components.) **(Compositions given are typical values, not specifications.)**

Component Name	CAS No.	Composition Amount (Wt.)		Carcinogen ###
Propylene Glycol	57-55-6	> 99	%	N/P

###1=U.S. National Toxicological Program 2=International Agency for Research on Cancer 3=U.S. Occupational Health and Safety Administration 4=American Conference of Governmental Industrial Hygienists 9=Other N/P=No Applicable Information Found

10. Component Health Hazards

Component	Component Health Hazards
Propylene Glycol	Slight eye irritant

11. Additional Toxicological Information

Component Name/Comments

Propylene Glycol
High concentrations of Propylene Glycol in water when held in contact with human skin under closed conditions have been reported to cause skin irritation (Cosmetics and Toiletries 99:83-91,1984). The authors attribute the observations to a sweat retention reaction by skin. No reactions were observed in open patch tests with human subjects. One literature report indicates rare eczematous skin reactions and even more rarely an allergic skin reaction from exposure to Propylene Glycol (Anderson and Starr, Hautzart 33 (1) 1982).

Material
No additional toxicology information is available for this material.

12. Physical and Chemical Data

Boiling Point ~ 188°C (at 760 mm Hg)	**Viscosity** ~ 46 mPa-s (at 25°C) (Brookfield)	**Dry Point** ~ 190°C
Freezing Point ~ -60°C	**Vapor Pressure** < .1 mm Hg (at 25°C)	**Volatile Characteristics** Slight
Density ~ 1040 kg/m3 (at 25°C)	**Vapor Specific Gravity** ~ 2.6 (Air = 1.0 at 15 - 20°C)	**Solubility in Water** Complete (In All Proportions)
pH Not applicable	**Hazardous Polymerization** Not expected to occur	**Stability** Stable

Other Chemical Reactivity	Reacts with strong oxidizing agents
Other Physical and Chemical Properties	Hygroscopic
Appearance and Odor	Clear, colorless; Slightly viscous liquid; Little or no odor
Conditions to Avoid	High temperatures, oxidizing conditions
Materials to Avoid	Strong oxidizing agents
Hazardous Decomposition Products	Incomplete combustion may produce carbon monoxide and other toxic gases

13. Hazards Rating Information

National Fire Protection Association
Health = 0 Flammability = 1 Reactivity = 0 Special Hazard - None
Ratings have been based on available component information from the National Fire Protection Association.

National Paint and Coatings Association

Hazardous Material Information System (HMIS)
Health = 0 Flammability = 1 Reactivity = 0
Ratings have been generated according to criteria specified in the National Paint and Coatings Association Implementation Manual based on component information available.

14. Additional Precautions

Handling and Storage Procedures
Hygroscopic. Use dry nitrogen or low dew point air for tank padding. Keep drums tightly closed to prevent contamination. Store at 20-30°C.

Decontamination Procedures
Isolate, vent, drain, wash and purge systems or equipment before maintenance or repair. Wear recommended personal protective equipment. Observe precautions pertaining to confined space entry.

15. Label Information

Source/Distributor
Lyondell Asia Pacific, Ltd.
41st Floor, The Lee Gardens,
33 Hysan Avenue
Causeway Bay
Hong Kong

Telephone Numbers:
EMERGENCY
(886) 933-635-556
CUSTOMER SERVICE
(852) 2882-2668

Other Company Names PG

Signal Word CAUTION

Use Statement
For industrial use only
Keep out of reach of children

Physical Hazards
Combustible
Aqueous solutions may produce flammable vapors

Health Hazards
Eye irritant

Precautionary Measures
Do not handle near heat, sparks, or open flame
Avoid contact with eyes
Use with adequate ventilation
Keep container tightly closed when not in use
Store in dry, clean area

IMO Proper Shipping Name Not regulated
IATA Proper Shipping Name Not regulated

Transport Hazard Classification

ADR/VLG	Not regulated	**RID/VSG**	Not regulated	**IMO**	Not regulated
IATA	Not regulated	**ADNR/VBG**	Not regulated		

UN/NA ID No. Not Classified

Component Name	CAS No.	Composition Amount (Wt.)	EINECS No.
Propylene Glycol	57-55-6	> 99 %	200-338-0

Instructions: **In case of fire, use:** Alcohol type foam; CO_2; Dry chemical

First Aid:

Inhalation Not expected to present a significant inhalation hazard under anticipated conditions of normal use.

Eye Contact In case of eye contact, immediately rinse with clean water for 20-30 minutes. Retract eyelids often. Obtain emergency medical attention if pain, blinking, tears or redness persist.

Skin Contact Not expected to present a significant skin hazard under anticipated conditions of normal use.

Ingestion Not expected to present a significant ingestion hazard under anticipated conditions of normal use.

In case of spill: May contaminate water supplies/pollute public waters. Slippery walking/spread granular cover or soak up. On water, may biodegrade. Contain/collect rapidly to minimize dispersion. Report per regulatory requirements.

Protective Equipment

Respiratory No special respiratory protection is recommended under anticipated conditions of normal use with adequate ventilation.

Eye Chemical splash goggles and/or face shield should be worn.

Skin No special clothing normally required. Where use can result in skin contact, wash thoroughly before eating, drinking, smoking, or leaving work.

Label No.: LP000029 **Version No.:** 2 **Date:** 27 January 1999

16. General Comments

General Comments
No data available.

Other Comments
No additional information available.

Note Qualifications:	= - Equal	~ - Approximately	N/P=No Applicable Information Found
	< - Less Than	UK - Unknown	N/AP=Not Applicable
	> - Greater Than	## - Trace	N/DA=No Data Available

Print Date 23 March 1999

Document Status APPROVED

SAFETY DATA SHEET

01 - IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND THE COMPANY/UNDERTAKING

PRODUCT NAME	FORANE 407C
SDS No.	01965
SUPPLIER	ELF ATOCHEM D. FLUORES ET OXYGENES Cours Michelet - La Défense 10 92091 PARIS LA DEFENSE CEDEX FRANCE Téléphone : 01 49 00 80 80 Télécopie : 01 49 00 83 96
Emergency telephone number	33 1 49 00 80 80

02 - COMPOSITION / INFORMATION ON INGREDIENTS

CHEMICAL NATURE OF THE PREPARATION	MIXTURE BASED ON : FORANE 32 (DIFLUOROMETHANE) CAS : 75-10-5 EINECS : 200-839-4 (*) FORANE 125 (PENTAFLUOROETHANE) CAS : 354-33-6 EINECS : 206-557-8 (*) FORANE 134a (1,1,1,2-TETRAFLUOROETHANE) CAS : 811-97-2 EINECS : 212-377-8 (*)

03 - HAZARDS IDENTIFICATION

MOST IMPORTANT HAZARDS	-
PHYSICAL AND CHEMICAL HAZARDS	Thermal decomposition giving toxic and corrosive products.

04 - FIRST AID MEASURES

GENERAL ADVICE	-
INHALATION (*)	Move to fresh air. Oxygen or artificial respiration if needed. In case of persistent problems : (*) Consult a doctor. (*)
SKIN CONTACT	Frostbite : treat as thermal burns.
EYE CONTACT	Wash immediately, abundantly and thoroughly with water. If irritation persists, consult an ophthalmologist.
PROTECTION OF FIRST-AIDERS (*)	In case of insufficient ventilation, wear suitable respiratory equipment. (*)
INFORMATION FOR DOCTORS	Do not administer catecholamines (because of the cardiac effect caused by the product)

05 - FIRE-FIGHTING MEASURES

SPECIFIC HAZARDS	Thermal decomposition giving toxic and corrosive products. Hydrogen fluoride

	Carbon oxides One of the components of this preparation gives flammable mixtures with air.
SPECIFIC METHODS	Cool containers / tanks with water spray. Prohibit all sources of sparks and ignition - Do not smoke.
SPECIAL PROTECTIVE EQUIPMENT FOR FIREFIGHTERS	Wear a self-contained breathing apparatus and protective suit.

06 - ACCIDENTAL RELEASE MEASURES

PERSONAL PROTECTION	Avoid contact with skin and eyes and inhalation of vapours. Wear personal protective equipment. In enclosed areas : ventilate or wear a self-contained breathing apparatus (risk of anoxia). Do not smoke.
ENVIRONMENTAL PROTECTION	Minimize as much as possible discharge into the environment

07 - HANDLING AND STORAGE

Technical measures/Precautions	Storage and handling precautions applicable to products : GAS UNDER PRESSURE Ensure appropriate exhaust and ventilation at machinery.
Safe handling advice	Prohibit ignition sources and contact with hot surfaces - DO NOT SMOKE.
Technical measures/Storage conditions	Store at ambient temperature in the original container. Keep away from naked flames, hot surfaces and sources of ignition. Keep in a cool, well-ventilated place. Protect full containers from sources of heat to avoid overpressurization.
Recommended	Ordinary steel
To be avoided	Alloys containing more than 2% of magnesium. Plastic materials

08 - EXPOSURE CONTROLS / PERSONAL PROTECTION

PROTECTIVE PROVISIONS	Ensure sufficient air exchange and/or exhaust in work areas.
CONTROL PARAMETERS	-
Exposure limits	No limit value F-USA FORANE 134a Value recommended by the "Comité Valeur limite d'exposition" of ELF ATOCHEM: VME = 1000 ppm - FORANE 32 Value proposed by the "Comité Valeur limite d'exposition" of ELF ATOCHEM : VME = 1000 ppm. Forane 125 Value proposed by the "Comité Valeur limite d'exposition" of ELF ATOCHEM : VME = 1000 ppm.
PERSONAL PROTECTION EQUIPMENT	-
Respiratory protection	In case of insufficient ventilation, wear suitable respiratory equipment.
Hand protection	Gloves
Eye protection	Safety glasses.

09 - PHYSICAL AND CHEMICAL PROPERTIES

PHYSICAL STATE (20°C)	liquefied gas
COLOUR	colourless
ODOUR	Slightly ether-like
pH	Not applicable
BOILING POINT/RANGE	-42.4 °C
FLASH POINT	No flash point (in the test conditions)
VAPOUR PRESSURE	(25°C) : 1.13 MPa (11.3 bar) (50°C) : 2.11 MPa (21.1 bar) (70°C) : 3.26 MPa (32.6 bar)
VAPOUR DENSITY	At the boiling point : 4.54 kg/m3
DENSITY	(25°C) : 1133 kg/m3 (50°C) : 1004 kg/m3 (70°C) : 861 kg/m3
PARTITION COEFFICIENT (n-octanol/water)	log Pow = 0.21 (Forane 32) - log Pow = 1.48 (Forane 125) - log Pow = 1.06 (Forane 134a)
OTHER DATA	Forane 134a : Does not dissociate in water Henry's constant : 3.09E5 Pa m3/mole (Forane 125) - : 1.55E5Pa m3/mole Critical temperature: Tc=89°C Critical pressure: Pc=4.64 MPa (46.4 bar)

10 - STABILITY AND REACTIVITY

CONDITIONS TO AVOID	Avoid contact with flames and red hot metallic surfaces.
HAZARDOUS DECOMPOSITION PRODUCTS	Thermal decomposition into toxic products containing fluorine Hydrogen fluoride (hydrofluoric acid) Carbon oxides
FURTHER INFORMATION	The product is stable under normal handling and storage conditions.

11 - TOXICOLOGICAL INFORMATION

ACUTE TOXICITY	-
Inhalation	Experimental effects on animals : FORANE 134a, FORANE 32, FORANE 125 Practically not harmful by inhalation. No mortality in rat at 500 000 ppm / 4h. As with other volatile aliphatic halogenated compounds, through vapour accumulation and/or inhalation of large quantities, the product can cause : Loss of consciousness and cardiac disorders aggravated by stress and lack of oxygen: risk of mortality.
LOCAL EFFECTS	-
Skin-contact	Ejection of liquefied gas : frostbite possible.
CHRONIC TOXICITY	FORANE 134a, FORANE 32, FORANE 125 Studies of prolonged inhalation in animals have not shown sub-chronic toxic effects. (rat /3 month(s)/Inhalation : 50 000 ppm)
SPECIFIC EFFECTS	GENOTOXICITY : According to available experimental data :

FORANE 134a, FORANE 32, FORANE 125
Not genotoxic
CARCINOGENICITY :
FORANE 134a
Experimentation on animals has not shown clear evidence of carcinogenic effect.
(rat/Inhalation - oral route)
REPRODUCTIVE TOXICITY :
Foetal development :
FORANE 134a, FORANE 32, FORANE 125
According to available experimental data :
Absence of toxic effects for foetal development
(Inhalation/rat - rabbit)
Fertility :
According to limited available data in animals :
FORANE 134a
Absence of toxic effects on fertility
(mouse/Inhalation)

12 - ECOLOGICAL INFORMATION

- SUBSTANCE CONCERNED	FORANE 32
PERSISTENCE/DEGRADABILITY	-
In water	Not readily biodegradable : 5% after 28d
BIOACCUMULATION	Practically not bioaccumulable : log Pow = 0.21 (measured)
- SUBSTANCE CONCERNED	FORANE 125
MOBILITY	Rapid evaporation : t½ life = 3.2h (estimated)
PERSISTENCE/DEGRADABILITY	-
In water	Not readily biodegradable 5% after 28d
In air	Degradation in the troposphere : t½ life = 28.3y (estimated) Ozone depletion potential : ODP (R-11 = 1) = 0 Global warming potential (GWP) : (HGWP) = 0.58
In soils and sediments	Slight adsorption : log Koc = 1.3 - 1.7
BIOACCUMULATION	Practically not bioaccumulable log Pow = 1.48
- SUBSTANCE CONCERNED	FORANE 134a
MOBILITY	Rapid evaporation : t½ life = 3h (estimated)
PERSISTENCE/DEGRADABILITY	-
In water	Not readily biodegradable : 3% after 28d
In air	Degradation in the atmosphere : 3 % after 28d Ozone depletion potential : ODP (R-11 = 1) = 0 Global warming potential (GWP) = 0.26
BIOACCUMULATION	Practically not bioaccumulable : log Pow = 1.06

13 - DISPOSAL CONSIDERATIONS

DISPOSAL OF PRODUCT	Recycle or incinerate at an approved waste disposal site only.

14 - TRANSPORT INFORMATION (*)

UN Number (*)	3340 (*)

ADR/RID	Class : 2 Item (letter) : 2°A
Prescriptions (*)	Labels : 2 H.I. Nr/ID Nr : 20/3340 (*)
IMDG (*)	Class : 2.2 UN Nr (IMDG) : 3340 (*)
Prescriptions	Labels : 2.2 (*)
IATA (*)	Class : 2.2 UN Nr (IATA) or ID Nr : 3340 (*)
Prescriptions	Labels : 2.2
	Consult ELF ATOCHEM's safety department for any further information

15 - REGULATORY INFORMATION

EEC DIRECTIVE	-
SAFETY DATA SHEETS	D. 91/155/EEC amended by D.93/112/EEC : Dangerous substances and preparations
EC CLASSIFICATION / LABELLING	-
DANGEROUS PREPARATIONS	D. 88/379/EEC amended by D. 93/18/EEC (3rd ATP) Not classified as dangerous
SUBSTANCES DAMAGING TO THE OZONE LAYER (*)	EC Regulation N° 3093/94 of 15.12.94. (*)
INVENTORIES (*)	EINECS (EU) : conforms (*) TSCA (USA) : conforms

16 - OTHER INFORMATION

RECOMMENDED USES	Refrigerant
BIBLIOGRAPHY REFERENCES	Encyclopédie des gaz (Air Liquide - Ed.1976 – ELSEVIER AMSTERDAM)

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This information applies to the PRODUCT AS SUCH and conforming to specifications of ELF ATOCHEM.
In case of formulations or mixtures, it is necessary to ascertain that a new danger will not appear.
The information contained is based on our knowledge of the product, at the date of publishing and it is given quite sincerely. However the revision of some data is in progress.
Users are advised of possible additional hazards when the product is used in applications for which it was not intended. This sheet shall only be used and reproduced for prevention and security
purposes.
The references to legislative, regulatory and codes of practice documents cannot be considered as exhaustive.
It is the responsibility of the person receiving the product to refer to the totality of the official documents concerning the use, the possession and the handling of the product.
It is also the responsibility of the handlers of the product to pass on to any subsequent persons who will come into contact with the product (usage, storage, cleaning of containers, other processes)
the totality of the information contained within this safety data sheet and necessary for safety at work, the protection of health and the protection of environment.

The (*) indicate the changes made with respect to the previous version.

End of document.
Number of page(s): 6